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An Optimized and Interpretable Deep Learning Framework for Brain Tumor Classification

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Contents

1. Introduction **4**
1.1 Background 4
1.2 Motivation 5
1.3 Objectives and Research Questions 6
1.4 Contributions 7

2. Literature Review **9**
2.1 Deep Learning for Brain Tumor Classification 9
2.2 Reference Pipeline and Comparative Studies 9
2.3 Attention Mechanisms in Deep Neural Networks 10
2.4 Explainable AI in Medical Imaging 11
2.5 Research Gaps and Motivation for This Work 13
2.6 Beyond Post-hoc Explainability: Broader Perspectives on XAI 15

3. Methodology **16**
3.1 Dataset Description 16
3.2 Data Preprocessing and Augmentation 18
3.3 Model Architecture 21
3.4 Optimization and Training Setup 24
3.5 Evaluation Metrics 27
3.6 Explainable AI Framework 30
3.7 Methodology Summary 32

4. Experiments and Results **33**
4.1 Overview 33
4.2 Baseline Model (EfficientNetV2-S with Adam) 34
4.3 Optimized Model (AdamW + Tuned Augmentation) 36
4.4 Attention Mechanisms (ECA, GAM, and CBAM) 39
4.5 Optimized Model with Improved Preprocessing 42
4.6 Optimized Model with Label Smoothing 50
4.7 Overall Comparison and Findings 52
4.8 Explainability Visualization of the Best Model 53

5. Discussion **56**
5.1 Overview 56
5.2 Impact of Optimization and Regularization 56
5.3 Why Attention Mechanisms Failed to Improve Performance 57
5.4 The Role of Preprocessing and Data Integrity 57
5.5 The Accuracy–Interpretability Trade-Off (Label Smoothing Case) 58
5.6 General Insights and Implications 59
6. Conclusion and Future Work **60**
6.1 Conclusion 60
6.2 Future Work 61

Abstract

Artificial intelligence has become an integral part of many applied fields, including medical diagnostics. One of the key challenges in this field is the automatic detection of brain tumors in magnetic resonance imaging (MRI) images. Deep convolutional neural networks have demonstrated high efficiency in this task, achieving impressive accuracy rates. However, the practical implementation of such models in clinical settings remains limited by their low interpretability. This thesis proposes an optimized and interpretable deep learning model based on the EfficientNetV2-S architecture for multi-class brain tumor classification, focusing on achieving a balance between high accuracy and decision transparency.

Several experiments were conducted with various architectural and training configurations, including models with channel- and spatial-attention modules (ECA, GAM, CBAM), an improved data preprocessing step, and the use of label smoothing. The optimized EfficientNetV2-S configuration using AdamW, tuned learning rates, and enhanced data augmentation achieved a test accuracy of 0.989, outperforming the baseline model (0.984).

Adding attention mechanisms resulted in a decrease in accuracy (≈0.977–0.980), indicating their redundancy due to the built-in squeeze-and-excitation mechanisms. The use of improved preprocessing restored both the accuracy and interpretability of the model, providing more balanced results across classes.

Interpretability was assessed using Faster Score-CAM, Grad-CAM++, Grad-CAM, and SmoothGrad, which consistently identified clinically significant tumor regions. A slight tradeoff between accuracy and interpretability was found: label smoothing slightly increased accuracy (to 0.99) but reduced the clarity of heatmaps.

Overall, the results demonstrate that high accuracy and interpretability can coexist. The optimized EfficientNetV2-S represents a robust and explainable model for classifying brain tumor MRI images and contributes to the development of transparent and trustworthy AI systems in healthcare.

1. Introduction

1.1 Background

Brain tumors are life-threatening neurological diseases caused by abnormal growth of brain cells. Early detection is essential for treatment and greatly increases the chances of survival. MRI is the gold standard for visualizing brain anatomy, revealing the size, shape, and location of brain tumors. Manual MRI interpretation is time-consuming and also heavily relies on the skills of the radiologist. With the growth of data volumes, ensuring consistent and timely diagnosis using older methods has become increasingly difficult.

In recent years, with the rapid development of deep learning, convolutional neural networks (CNNs) have made significant advances in medical image analysis. CNNs learn from data and automatically extract features, outperforming manual selection. In brain classification, CNNs are capable of detecting and identifying brain tumors: glioma, meningioma, pituitary tumor, and no tumor with high accuracy. Transfer learning accelerates this process by using a trained feature extractor on massive datasets, which in turn significantly speeds up training time. Although CNNs boast high accuracy, they are black-box models. This means we don't know how the model made its decisions. In medicine, opacity, reliability, accountability, and ethical deployment are important. Not only is numerical accuracy important, but explainability and trustworthiness are also crucial. People need to understand how the model reached its decision and that decisions were made based on pathologically significant regions, not on image artifacts.

For this purpose, explainable artificial intelligence exists as a bridge between accuracy and interpretability. I used GradCAM, GradCAM++, Faster-ScoreCAM, and SmoothGrad, which create heatmaps that highlight the most significant regions for disease detection. Combining CNNs and XAI not only achieves high speed but also enhances trustworthiness and human-AI interaction.

However, not everything is so smooth; many studies indicate that interpretability can often reduce accuracy, leading to a tradeoff between accuracy and interpretability. Therefore, a balance between model complexity, predictive power, and interpretability is important.

We used the EfficientNetV2-S framework, which combines efficiency with superior feature extraction through compound scaling and built-in squeeze-and-excitation mechanisms. However, while EfficientNetV2-S already effectively captures multi-scale visual information, its interpretability in medical domains, especially in brain MRI classification, remains underexplored. In this work, we attempt to train EfficientNetV2-S while maintaining a balance between accuracy and interpretability.

1.2 Motivation

Recent advances in deep convolutional neural networks have led to significant progress in medical image analysis, but their integration into clinical practice remains limited. One of the main reasons is that high-accuracy models are often difficult to interpret, which makes them hard to validate and trust in healthcare settings. Radiologists and physicians require not only accurate predictions but also an understanding of how model decisions are made. Therefore, there is a growing need for models that are both accurate and interpretable, so that artificial intelligence systems can effectively support medical reasoning.

The main motivation of this project is the intersection of two goals, namely state-of-the-art prediction accuracy together with visualization through deep XAI. In the previous work, two models were trained, XceptionNet and EfficientNetV2-S, which showed good classification of affected skin areas. From this work, experience was obtained in preprocessing, augmentation, and explainability through Faster Score-CAM and SmoothGrad. However, when transferring the application domain to brain tumor MRI, problems appeared: higher intra-class variability, subtle tumor boundaries, and more pronounced class imbalance.

The EfficientNetV2-S architecture was chosen in this study because of the balance between accuracy, efficiency, and resistance to overfitting. Its compound scaling strategy optimizes network depth, width, and resolution, while the embedded squeeze-and-excitation (SE) layers already provide a certain level of channel-wise attention. Because of this, a question appeared: will additional attention mechanisms improve feature extraction, or will the model start to degrade due to an excess of attention?

To answer this question, attention mechanisms such as ECA, GAM, and CBAM were used. They showed success in some papers where they improved feature localization and discriminative learning. However, in our project they showed uncertain results. Another key goal is how preprocessing and augmentation affect not only prediction accuracy, but also the clarity and stability of the XAI heatmap.

There are also strategies that not only increase prediction accuracy, but also worsen the sharpness of the explanation. A clear example is label smoothing, which increases accuracy and reduces overconfidence, but blurs the heatmap. This contradiction between performance optimization and explainable transparency defines the main motivation of this work.

In the end, the goal is to create a framework for brain tumor detection with high accuracy and with visualization of decisions that can work in clinical conditions. By creating multiple code variants, the study aims to find a balance between high accuracy and stable, meaningful interpretability, thereby advancing the development of reliable AI for medical diagnosis.

1.3 Objectives and Research Questions

The main aim of this research is to design, implement, and evaluate an optimized and interpretable deep learning framework for multiclass brain tumor classification using MRI images. The proposed system aims to balance high predictive accuracy with clinical interpretability, so that model decisions are transparent, reproducible, and consistent with medical reasoning.

General Objective

To develop an EfficientNetV2-S–based deep neural network that achieves strong classification performance together with meaningful interpretability through explainable AI (XAI) visualization methods.

Specific Objectives

1. **Model Optimization**
   To improve the baseline EfficientNetV2-S architecture through better preprocessing, advanced data augmentation, and optimized hyperparameters (learning rate, batch size, optimizer type).
2. **Attention Mechanism Evaluation**
   To integrate and evaluate different attention modules, ECA (Efficient Channel Attention), GAM (Global Attention Module), and CBAM (Convolutional Block Attention Module), and analyze their effect on performance and interpretability within the EfficientNetV2-S backbone.
3. **Explainable AI Integration**
   To apply and compare explainable AI techniques (Grad-CAM, Grad-CAM++, Faster Score-CAM, SmoothGrad) as tools for interpreting and analyzing model behavior.
4. **Interpretability–Performance Trade-off Analysis**
   To study how optimization strategies such as label smoothing and regularization affect both quantitative metrics (accuracy, F1-score) and qualitative interpretability (saliency focus and clinical plausibility).

Research Questions

1. How can the EfficientNetV2-S architecture be optimized to improve both accuracy and training stability in brain tumor MRI classification?
2. Do attention mechanisms (ECA, GAM, CBAM) provide measurable improvements in performance or interpretability when added to EfficientNetV2-S, or do they introduce redundancy?
3. How do data preprocessing and augmentation techniques affect the model's generalization ability and the quality of XAI visualizations?
4. What is the nature of the accuracy–interpretability trade-off, and under which conditions can it be reduced or balanced?

This section outlines the main analytical direction of the thesis: each experimental variant, interpretability method, and evaluation step is designed to address these objectives and questions. The following chapters present the methodology, experiments, and results that together provide answers to them.

1.4 Contributions

This thesis contributes to the field of interpretable deep learning for medical imaging by presenting a systematic study that combines model optimization, explainability, and clinical relevance. The main contributions of this work are summarized as follows:

1. **An Optimized EfficientNetV2-S Framework for Brain Tumor MRI Classification**
   Developed and fine-tuned a deep learning model based on EfficientNetV2-S using AdamW optimization, learning rate scheduling, and improved data preprocessing and augmentation strategies.
   The resulting configuration achieved a test accuracy of 0.989, outperforming the baseline (0.984), while showing stable validation behavior and balanced class-wise performance.
2. **Comprehensive Evaluation of Attention Mechanisms**
   Implemented and evaluated ECA, GAM, and CBAM modules within the EfficientNetV2-S backbone.
   Performed a systematic comparison showing that, despite reported success in other computer vision tasks, these modules did not improve performance on MRI data and reduced interpretability, likely due to redundancy with EfficientNet's built-in squeeze-and-excitation layers.
   This result provides additional insight into the limitations of external attention modules when applied to attention-rich architectures in medical imaging.
3. **Improved Preprocessing Pipeline for Medical MRI Data**
   Designed an enhanced preprocessing workflow that includes brightness normalization, contrast correction, and balanced augmentation.

Showed that better input conditioning can lead to improved convergence and more balanced confusion matrices, even without changes to the model architecture.

4. **Systematic Analysis of the Accuracy–Interpretability Trade-off**
   Investigated the effects of label smoothing and regularization on both classification metrics and the quality of XAI visualizations.
   Observed that small gains in accuracy (0.990) may come at the cost of less focused saliency maps, confirming a measurable trade-off between performance optimization and interpretability.
5. **Integration and Comparative Validation of Explainable AI Techniques**
   Applied and compared Faster Score-CAM, Grad-CAM, Grad-CAM++, and SmoothGrad for visualizing model decisions.
   Demonstrated that the optimized model provides consistent tumor localization across all four classes, with the clearest and most focused explanations produced by Faster Score-CAM.
   Provided a qualitative XAI benchmark to support future reproducible research on interpretability.

Together, these contributions show that interpretability and accuracy are not necessarily conflicting goals. By systematically analyzing preprocessing, architectural choices, and explainability within a unified experimental framework, this work provides a solid basis for developing transparent and clinically reliable deep learning systems for brain tumor classification.

2. Literature Review

2.1 Deep Learning for Brain Tumor Classification

Deep learning has become the dominant paradigm in medical image analysis, providing automated diagnostic tools capable of learning complex spatial relationships directly from imaging data. Convolutional Neural Networks (CNNs) [22], in particular, have proven highly effective in brain tumor detection and classification tasks using magnetic resonance imaging (MRI). Their ability to extract hierarchical features from raw images enables more robust differentiation between tumor types such as glioma, meningioma, pituitary tumor, and no tumor.

While traditional machine learning approaches relied on manual feature engineering and domain-specific preprocessing, CNNs learn representations directly from data, achieving higher accuracy and generalization [24]. Transfer learning further accelerated this progress by allowing pre-trained architectures such as Xception, ResNet, and EfficientNet to be fine-tuned for medical imaging with limited datasets [23]. EfficientNet, introduced by Tan and Le, employs a compound scaling strategy that jointly balances network depth, width, and input resolution, enabling improved accuracy–efficiency trade-offs and strong transfer learning performance compared to conventional CNN architectures [16]. However, these models often remain black boxes, providing little insight into the decision-making process. As clinical adoption depends on explainability and reliability, recent research has increasingly focused on interpretable deep learning and Explainable Artificial Intelligence (XAI) techniques, including Grad-CAM, Grad-CAM++, Faster Score-CAM, and SmoothGrad to visualize which image regions influence model predictions. Early studies such as the work by Paul et al. demonstrated that convolutional neural networks can effectively classify brain tumors from MRI data, providing early evidence of the suitability of deep learning approaches for multi-class brain tumor classification [21].

2.2 Reference Pipeline and Comparative Studies

The proposed framework in this thesis builds upon and extends the methodology described by Mahmud et al. (2023) in their work "An Interpretable Deep Learning Approach for Skin Cancer Categorization", presented at the IEEE ICCIT 2023 Conference in Bangladesh [1].

Their study introduced a multi-model pipeline employing XceptionNet, EfficientNetV2-S, EfficientNetV2-M, and InceptionResNetV2, integrating Faster Score-CAM and SmoothGrad to improve interpretability [1]. Their results demonstrated that XceptionNet achieved the best performance (88.72 % accuracy) on the HAM10000 dataset, showing that explainable AI (XAI) can complement deep learning in medical image analysis.

That work served as an architectural and methodological foundation for the present research [1]. While Mahmud et al. focused on skin cancer, this thesis adapts and optimizes the same design principles for brain tumor MRI classification, a domain with greater spatial complexity and class variability. The fundamental ideas retained from Mahmud et al. (2023) include [1]:

- Use of EfficientNetV2-S as a strong and computationally efficient backbone.
- Integration of XAI techniques (Faster Score-CAM and SmoothGrad) to visualize model reasoning.
- Emphasis on balanced preprocessing, data augmentation, and hyperparameter optimization to improve generalization [25].

However, the present study expands on that baseline through several key innovations:

- Transition from skin lesion to brain tumor MRI classification (4 classes instead of 7).
- Introduction of AdamW optimization, fine-tuned learning rate scheduling, and advanced normalization for greater convergence stability.
- Integration and empirical analysis of attention modules (ECA, GAM, CBAM) to investigate whether additional feature-refinement layers enhance or degrade interpretability.
- A quantitative accuracy–interpretability analysis showing that certain regularization strategies (e.g., label smoothing) improve metrics slightly while dispersing heatmaps.

The importance of optimization strategies for training stability and generalization in medical imaging has also been highlighted by Mortazi et al., who showed that training dynamics can significantly influence model robustness beyond architecture choice [14].

For external benchmarking, the study refers to Disci et al. (2025, Cancers), who evaluated multiple transfer-learning architectures such as Xception, ResNet50, and DenseNet121 for MRI brain tumor classification, reporting up to 98.73 % accuracy [2]. Unlike those purely performance-oriented studies, this research emphasizes clinical transparency and explainable model behavior, establishing a balanced framework that aligns with the goals of trustworthy medical AI.

2.3 Attention Mechanisms in Deep Neural Networks

In parallel to optimization-based approaches, recent literature has explored the use of attention mechanisms to improve feature extraction and model focus.
Modules such as:

- Efficient Channel Attention (ECA) - introduces lightweight channel-wise weighting to emphasize discriminative features without heavy computation.

- Global Attention Module (GAM) - combines spatial and channel dependencies for global context awareness.
- Convolutional Block Attention Module (CBAM) - sequentially applies channel and spatial attention to refine feature maps.

These mechanisms were originally proposed to enhance CNNs like ResNet and DenseNet, and several studies reported accuracy improvements in natural image classification. For example, the Cluster Computing (2024) paper by Pacal et al. integrated ECA and GAM into EfficientNetV2, claiming a test accuracy of 99.76% on MRI data [3]. However, such results are often achieved under idealized settings and may not generalize to real-world variability.

In this thesis, these modules were experimentally tested within the EfficientNetV2-S architecture to verify whether attention integration genuinely improves performance in MRI-based classification. Empirical findings revealed the opposite: attention modules slightly reduced accuracy (≈0.977–0.980). This outcome suggests that EfficientNetV2-S, which already includes squeeze-and-excitation (SE) layers, may not benefit from additional external attention mechanisms. Instead, over-parameterization introduced unnecessary redundancy and weakened feature localization.

Therefore, while attention modules remain an influential design concept in CNN research, this study demonstrates that architectural optimization and preprocessing quality play a more significant role than adding extra attention layers when the base model already encodes efficient channel sensitivity.

2.4 Explainable AI in Medical Imaging

The rapid progress of deep learning in medical imaging has brought unprecedented diagnostic capabilities but has also intensified concerns over the opacity of neural networks. The lack of interpretability - often referred to as the “black-box” problem - presents a major obstacle for clinical acceptance. In domains where diagnostic accountability and patient safety are critical, healthcare professionals must be able to trust and verify the reasoning behind AI-driven predictions.

To address this, the field of Explainable Artificial Intelligence (XAI) has introduced a range of visualization and attribution techniques that reveal which image regions influence model decisions. Among the most widely adopted are Gradient-weighted Class Activation Mapping (Grad-CAM [17]), Grad-CAM++ [18], Score-CAM and its optimized variant Faster Score-CAM [19],which was specifically designed to reduce the computational overhead of activation-based explanations, alongside SmoothGrad [20], which enhances clarity by averaging gradient information across noisy input perturbations.

Grad-CAM and Grad-CAM++ provide gradient-based localization maps that highlight important regions corresponding to a model's predicted class. However, these methods are sensitive to noisy gradients and may produce unstable results when applied to deep architectures such as EfficientNet or Inception-based networks. In contrast, Score-CAM and Faster Score-CAM are gradient-free, relying instead on activation-based weighting, which produces smoother and more robust heatmaps. Faster Score-CAM, in particular, significantly reduces computational cost by using selective channel sampling, making it suitable for large-scale evaluation and real-time visual analysis.

From a clinical perspective, Chaddad et al. (2023) emphasized that explainability in medical imaging should be tailored to clinical tasks and user needs, as clinicians value explanations that enhance trust and reliability rather than generic visualizations [10].

In the study by Mahmud et al. (2023), Faster Score-CAM and SmoothGrad were successfully applied to improve the interpretability of XceptionNet and EfficientNetV2 models for skin cancer diagnosis [1]. These methods enabled the visualization of regions corresponding to malignant lesions, providing clinicians with more confidence in the model's predictions. Building on this foundation, the present research applies the same XAI techniques to MRI-based brain tumor classification, where anatomical complexity and texture variation pose additional challenges for explainability.

Furthermore, this thesis integrates XAI not merely as a visualization tool but as a quantitative diagnostic component - assessing the clarity, consistency, and localization accuracy of heatmaps across model variants. This dual use of XAI for both evaluation and interpretation supports the broader aim of developing trustworthy and clinically transparent AI systems.

Similar concerns regarding the limitations of post-hoc explainability in brain imaging were raised by Zeineldin et al. (2022), who specifically studied explainability in deep neural networks for MRI-based brain tumor classification and segmentation [4]. The authors proposed NeuroXAI, a modular framework that integrates multiple XAI methods and enables comparative analysis of different explanation techniques within the same model. Their results showed that different XAI methods often produce noticeably different localization patterns, even when applied to the same prediction, highlighting the risk of relying on a single saliency approach. Moreover, the study emphasized the importance of analyzing internal feature representations, rather than only final-layer heatmaps, as superficial visualizations may be insufficient to justify clinical decisions. This work reinforces the view that explainability is not an auxiliary add-on, but a prerequisite for trustworthy deployment of deep learning systems in brain tumor analysis.

Similar concerns about black-box behavior and the risk of misleading explanations were also highlighted by Hassija et al. (2023), who emphasized the persistent tension between model accuracy and human interpretability, particularly in high-stakes domains such as healthcare [9].

At a broader level, the theoretical foundations and limitations of explainable artificial intelligence have been systematically analyzed by Angelov et al. (2021), who provided an analytical review of XAI methods across different model classes and application domains [5]. The authors emphasized that many popular post-hoc explanation techniques, while visually intuitive, do not guarantee faithful representation of the underlying decision process and may lead to overconfidence if used without proper validation.

In particular, the review highlighted the lack of standardized evaluation criteria for explanation quality and the frequent mismatch between human interpretability and model fidelity. These observations are especially relevant in medical imaging, where explanations are often used to justify high-stakes clinical decisions. Angelov et al. further argued that explainability should be treated as a system-level requirement linked to trust, accountability, and regulatory compliance, rather than as a purely technical visualization tool.

This perspective aligns with the motivation of the present work, which treats XAI not as an optional post-hoc add-on, but as an integral component for analyzing model behavior, stability, and reliability in MRI-based brain tumor classification.

2.5 Research Gaps and Motivation for This Work

Despite numerous advancements in brain tumor classification and model interpretability, several research gaps remain that justify the direction and novelty of this work.

1. **Limited Generalization Across Modalities and Architectures.**
   Most prior research focuses on either performance optimization or visualization, rarely integrating both within a single, reproducible framework. Models achieving near-perfect accuracy often rely on small, homogeneous datasets, limiting their clinical relevance. Moreover, findings from attention-based architectures (ECA, CBAM, GAM) are inconsistent and seldom verified on diverse MRI datasets. This limitation is further compounded by the fact that high test accuracy does not necessarily imply reliable model behavior, as shown by Paschali et al. (2018), who demonstrated that models with similar generalization performance can exhibit substantially different robustness characteristics [15].
2. **Redundancy of External Attention Modules in Advanced Architectures.**
   While attention mechanisms have improved performance in lightweight CNNs such as ResNet or VGG, their usefulness in newer architectures like EfficientNetV2-S, which already includes internal squeeze-and-excitation layers, remains questionable. Prior studies claimed marginal gains using ECA or GAM; however, empirical results in this research reveal that these modules may introduce redundancy, slightly reducing performance.

3. **Insufficient Integration of Explainable AI in Quantitative Evaluation.**
   Most previous works (including those reporting top accuracies around 98–99%) employ Grad-CAM purely for post-hoc visualization. Few studies systematically analyze how optimization techniques or architectural changes affect both numerical performance and interpretability quality, despite prior analytical reviews highlighting this gap as a core limitation of current XAI practice [5]. This leaves an unexplored research space where XAI can serve as an analytic tool - not just a presentation method. Similar lifecycle-oriented limitations were also discussed by Dwivedi et al. (2023), who argued that explainability should be embedded across data preparation, model training, and evaluation stages, rather than being treated as a purely post-hoc visualization step [12]. Ali et al. (2023) further emphasized that explainability, while essential, represents only one component of trustworthy AI and must be considered alongside robustness, fairness, and the evaluation of explanations, particularly in high-stakes domains such as healthcare [13].
4. **Lack of Stability and Interpretability Balance.**
   While Mahmud et al. (2023) successfully demonstrated the feasibility of integrating Faster Score-CAM and SmoothGrad into a dermatological diagnostic pipeline [1], their approach was not optimized for stability in MRI-based classification tasks. The present work extends that idea, focusing on training stability, class balance, and interpretability coherence across variants.
5. **Missing Discussion of the Accuracy–Interpretability Trade-off.**
   Although widely acknowledged in XAI literature, this trade-off has not been empirically quantified in most CNN-based studies for medical imaging. Through systematic experimentation, this thesis provides concrete evidence that marginal accuracy improvements - for instance, from label smoothing (0.989 → 0.990) — can result in weaker saliency focus, highlighting the importance of balanced optimization.

This broader motivation aligns with the Healthcare 5.0 perspective discussed by Saraswat et al. (2022), which treats explainability as a core requirement for trustworthy clinical AI systems rather than a secondary feature [11].

2.6 Beyond Post-hoc Explainability: Broader Perspectives on XAI

While most explainable AI research in medical imaging concentrates on discriminative models such as classifiers and segmenters, recent work suggests that explainability challenges extend beyond post-hoc visual attribution alone. In this context, Sun et al. (2022) examined explainability in generative AI systems and argued that traditional XAI approaches are often insufficient for models producing open-ended outputs rather than single predictions [6].

Adopting a human-centered explainable AI (HCXAI) perspective, the authors emphasized that explanations should be aligned with users' understanding needs instead of being purely model-driven. Their findings highlight that users require information about model limitations, uncertainty, and context, in addition to visual explanations. Although this study was conducted in the domain of software engineering, its core insight is also relevant for medical AI systems, where clinicians must form reliable mental models of AI behavior rather than rely solely on saliency maps.

At a more general level, similar limitations of current XAI practice were identified by Saeed and Omlin (2023) in their systematic meta-survey of explainable AI research [8]. The authors highlighted persistent challenges such as the lack of standardized evaluation metrics, ambiguity between interpretability and explainability, and trade-offs between transparency and predictive performance. Importantly, they argued that explainability should be considered across the entire machine learning lifecycle, rather than being treated as a purely post-hoc analysis step.

More recent work has further extended this discussion to large-scale generative systems. Schneider (2024) introduced the concept of Explainable Generative AI (GenXAI), emphasizing that explainability should be treated as a system-level property rather than a single visualization technique [7]. Although this perspective primarily targets generative models, it reinforces a broader insight that is also relevant for medical imaging: trustworthy AI systems require integrated and task-aware explainability, not isolated explanatory add-ons.

3. Methodology

3.1 Dataset Description

This research is based on the Brain Tumor MRI dataset publicly available on Kaggle, which provides a well-structured benchmark for supervised brain tumor classification tasks.

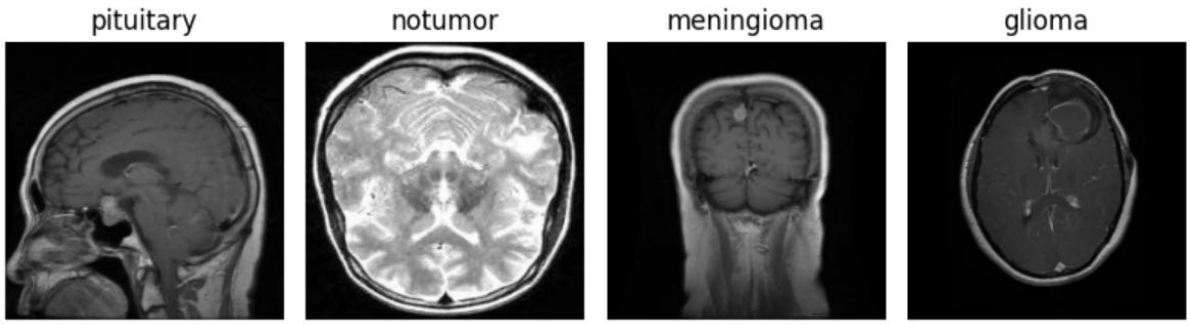

Figure 3.1: Representative MRI samples for each diagnostic category in the Brain Tumor MRI dataset (pituitary tumor, no tumor, meningioma, and glioma).

The dataset consists of 6,597 high-resolution MRI images divided into four diagnostic categories: glioma, meningioma, pituitary tumor, and no tumor. Each image corresponds to a 2D slice of brain MRI, representing different imaging orientations (axial, sagittal, coronal) and varying intensity levels due to scanner type, acquisition parameters, and patient anatomy. Representative examples of MRI images for each diagnostic category are shown in Figure 3.1.

3.1.1 Data Cleaning and Integrity Verification

To ensure data integrity and eliminate redundancy, all image files were programmatically scanned and validated using SHA-1 hashing. Duplicate files were automatically detected based on pixel-level content similarity and removed. In total, 426 duplicate images were identified and excluded, reducing the dataset to 6,597 unique and non-overlapping samples, guaranteeing that each training and testing sample contributes unique information.

3.1.2 Class Composition

Table 3.1: Class composition of the Brain Tumor MRI dataset after duplicate removal.

Tumor Class	Sample Count	Percentage
Pituitary tumor	1,740	26.4%
No tumor	1,706	25.9%
Glioma	1,620	24.6%
Meningioma	1,531	23.2%
Total	6,597	100%

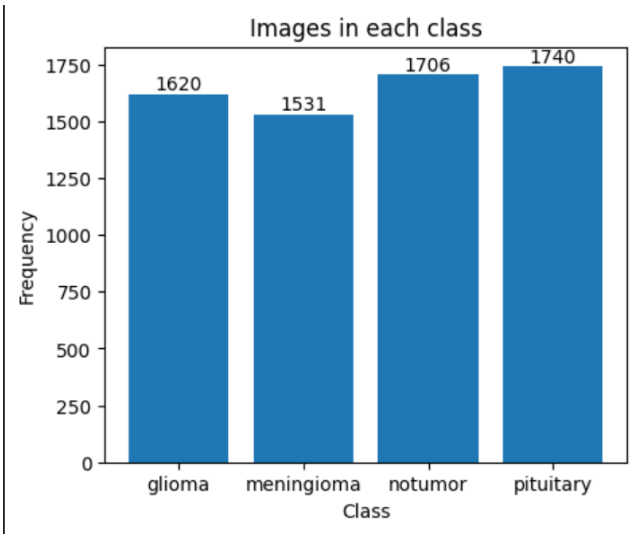

Figure 3.2: Class distribution of the Brain Tumor MRI dataset after duplicate removal.

Figure 3.2 visually confirms the relatively balanced class distribution reported in Table 3.1, indicating minimal class imbalance.

3.1.3 Data Splitting Strategy

For experimental reproducibility, the cleaned dataset was split using stratified sampling into three subsets:

- Training set (80%) – 5,343 images
- Validation set (10%) – 594 images
- Testing set (10%) – 660 images

Stratification ensured that each subset preserved the overall class ratio, preventing data leakage and guaranteeing fair evaluation. The split was performed with a fixed random seed (42) for deterministic reproducibility.

3.1.4 Data Storage and Format

All preprocessed images were converted to NumPy arrays (.npy) to accelerate I/O operations during multiple experimental runs. Each image was resized to 224×224×3 pixels, converted from BGR (OpenCV format) to RGB, and stored alongside one-hot encoded labels representing the four tumor classes. The resulting data tensors had the following shapes:

- data.npy: (6,597, 224, 224, 3)
- labels.npy: (6,597, 4)

This uniform preprocessing step was applied to avoid confounding effects caused by input inconsistencies across experiments.

3.1.5 Clinical Relevance

Unlike many synthetic or artificially augmented medical datasets, this MRI collection represents real-world variability:

- Multiple MRI machines and acquisition protocols.
- Tumors of various sizes, intensities, and anatomical locations.
- Differences in slice orientation and illumination.

Such diversity makes this dataset particularly suitable for evaluating model robustness, generalization, and interpretability, which are critical for medical deployment.

3.2 Data Preprocessing and Augmentation

Preprocessing plays an important role in deep learning for medical imaging, especially for MRI data, which often suffer from non-uniform brightness, scanner noise, and limited dataset size. For this reason, a preprocessing and augmentation pipeline was implemented to improve data consistency, model generalization, and visualization with XAI methods.

3.2.1 Image Preprocessing Pipeline

The preprocessing phase was executed before model training and included the following operations:

1. **Image Loading and Standardization**
   All MRI images were loaded using OpenCV, converted to the RGB color space, and resized to 224 × 224 pixels. This step ensured compatibility with the EfficientNetV2-S input requirements and consistent spatial dimensions across the dataset.
2. **Normalization**
   Pixel intensities were scaled to the range [0, 1] using a rescaling factor (1/255) applied during data generation and model training.
3. **Contrast and Brightness Enhancement**
   A mild brightness adjustment was applied during data augmentation to improve robustness to illumination variability and scanner-dependent intensity differences.
4. **Duplicate Elimination**
   The SHA-1–based duplicate filtering step described in Section 3.1 was an essential part of preprocessing, ensuring that model evaluation remained fair and free from repeated images across splits.
5. **Label Encoding**
   Each of the four classes (glioma, meningioma, pituitary, notumor) was encoded using a LabelEncoder and converted into a one-hot vector (dimension = 4), supporting multiclass softmax classification.

The final cleaned and normalized dataset provided a uniform baseline for all subsequent experiments and variant comparisons.

3.2.2 Data Augmentation Strategy

Given that deep convolutional networks are prone to overfitting when trained on limited or homogeneous medical datasets, an online data augmentation strategy was employed using TensorFlow's ImageDataGenerator.

This approach increased the effective diversity of the training data while preserving diagnostically relevant tumor structures. The final online data augmentation strategy and its corresponding parameters are summarized in Table 3.2

Table 3.2: Data augmentation transformations and corresponding parameter ranges used during training.

Transformation	Range / Parameter	Purpose
Rotation	±22°	Simulate variability in patient head orientation
Width & Height Shift	±12%	Account for spatial displacement of brain structures
Shear	0.10	Improve robustness to geometric distortions
Zoom	±22% (≈ [0.78×, 1.22×])	Model tumors at different apparent scales
Brightness Range	[0.85, 1.15]	Adapt to scanner-dependent illumination differences
Horizontal Flip	Enabled	Increase anatomical diversity
Rescaling	1/255	Normalize pixel intensities during training

Each transformation was applied stochastically by the generator at runtime, ensuring that augmented samples differed across training iterations. This real-time augmentation strategy reduced memorization, improved generalization, and enhanced robustness to unseen MRI variations.

A visual inspection of randomly augmented samples was performed to confirm that tumor regions remained anatomically consistent and free from unrealistic distortions.

3.2.3 Enhanced Preprocessing Version

Following the initial baseline configuration, an improved preprocessing setup was explored as part of the model optimization process. This version incorporated refined brightness adjustment parameters and a more balanced application of data augmentation across classes, aiming to stabilize training dynamics and reduce class-wise performance discrepancies.

Qualitative analysis of input intensity distributions suggested a more uniform appearance across tumor classes after refinement, reducing bias toward specific tumor subtypes.

This enhanced preprocessing configuration was subsequently adopted in the optimized EfficientNetV2-S model. The impact of this refinement on classification performance and interpretability is analyzed in Chapter 4.

3.3 Model Architecture

The backbone of the proposed framework is EfficientNetV2-S, a modern convolutional neural network architecture known for its balance between accuracy, computational efficiency, and scalability.

EfficientNetV2 was introduced as an improved successor to the original EfficientNet family, combining compound scaling (balancing depth, width, and resolution) with progressive learning and fused-MBConv blocks that significantly reduce training time and memory consumption.

This section describes the adopted architecture, custom classifier head, training configuration, and experimental variants incorporating attention modules.

3.3.1 EfficientNetV2-S Backbone

The EfficientNetV2-S model was initialized with ImageNet pre-trained weights, using the Keras implementation from TensorFlow (tensorflow.keras.applications.efficientnet_v2), as shown in Figure 3.3. The top classification layers were removed (include_top=False), allowing adaptation to the four-class brain-tumor classification problem.

The resulting convolutional base extracts multi-scale features across successive blocks, each consisting of:

- Fused and depthwise separable convolutions,
- Batch normalization,
- Swish activation (SiLU), and
- Built-in Squeeze-and-Excitation (SE) layers that perform channel-wise attention to emphasize the most informative feature maps.

```
base_model = EfficientNetV2S(weights='imagenet',
                             include_top=False,
                             input_shape=(224, 224, 3))
```

Figure 3.3: EfficientNetV2-S convolutional backbone initialized with ImageNet pre-trained weights

This inherent SE mechanism makes EfficientNetV2-S already attention-aware, potentially limiting the benefit of external attention modules are an aspect later investigated in the experimental section.

3.3.2 Custom Classification Head

On top of the pre-trained convolutional base, a lightweight yet expressive custom head was constructed to adapt the model for the four output classes.
The head consists of the following layers:

1. Global Average Pooling (GAP) condenses spatial feature maps into a single feature vector per channel, preserving the global activation distribution.
2. Dense(256, ReLU) fully connected layer with 256 neurons, expanding the representational capacity and enabling non-linear feature combinations.
3. Batch Normalization stabilizes internal activations, accelerates convergence, and mitigates overfitting.
4. Dropout(0.5) randomly drops half of the neurons during training, improving generalization by preventing co-adaptation of features.
5. Dense(4, Softmax) output layer providing class probabilities for the four MRI categories (glioma, meningioma, pituitary, no tumor).

The complete architecture can be summarized as:

```
x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dense(256, activation='relu')(x)        # was 128
x = BatchNormalization()(x)                 # helps regularize and stabilize training
x = Dropout(0.5)(x)
predictions = Dense(4, activation='softmax')(x)
```

Figure 3.4: Custom classification head built on top of the EfficientNetV2-S backbone, consisting of Global Average Pooling, fully connected layers, batch normalization, dropout, and a Softmax output layer.

This design achieved an optimal trade-off between expressive capacity and computational efficiency. Compared to heavier backbones such as Xception or Inception-ResNetV2, EfficientNetV2-S offers a smaller parameter count and faster training while maintaining comparable or superior accuracy.

3.3.3 Optimization Strategy

The model was compiled using categorical cross-entropy loss suitable for multi-class problems and the AdamW optimizer, which combines the adaptive gradient mechanism of Adam with explicit weight-decay regularization. This modification effectively controls overfitting and improves generalization, particularly for high-capacity networks.

```
model.compile(
    optimizer=AdamW(learning_rate=5e-4, weight_decay=1e-4),  # AdamW instead of plain Adam
    loss='categorical_crossentropy',
    metrics=['accuracy']
)
```

Figure 3.5: Model compilation using the AdamW optimizer with categorical cross-entropy loss and explicit weight decay.

Key hyperparameters were as follows:

- Learning rate: 5×10^{-4} (decayed automatically via scheduler)
- Weight decay: 1×10^{-4}
- Batch size: 16
- Epochs: 30 (with early stopping at ~17)

Learning-Rate Scheduling and Early Stopping

Two main callbacks were used to stabilize training:

1. ReduceLROnPlateau - halved the learning rate (factor = 0.5) after 3 consecutive epochs without improvement in validation accuracy (minimum LR = 1×10^{-6}).
   This allowed fine-grained convergence in later stages.
2. EarlyStopping - terminated training after 7 stagnant epochs and restored the best model weights, preventing overfitting.
3. ModelCheckpoint - saved only the best-performing model (EfficientNetV2S_0511(2).h5) based on validation accuracy.

```
early_stop = EarlyStopping(
    monitor='val_accuracy',
    patience=7,           # prevents overfitting, restores best weights
    restore_best_weights=True,
    mode='max',
    verbose=1
)
```

Figure 3.6: Training callbacks configuration, including ReduceLROnPlateau, EarlyStopping, and ModelCheckpoint.

This configuration was designed to promote stable convergence and prevent overfitting during training, confirming that the model had reached near-optimal performance without signs of divergence.

3.3.4 Experimental Variants with Attention Modules

To explore whether additional attention mechanisms could further enhance feature discrimination, several architectural variants were implemented and tested:

1. ECA (Efficient Channel Attention) adds lightweight 1D convolution across channels to strengthen inter-channel dependencies without dimensional reduction.
2. GAM (Global Attention Module) fuses spatial and channel attention to globally refine feature maps.
3. CBAM (Convolutional Block Attention Module) sequentially applies channel and spatial attention submodules to focus on informative features.

Each module was inserted after the convolutional backbone and before the GAP layer, keeping the rest of the classifier head unchanged. These architectural variants were implemented to assess whether additional attention mechanisms could enhance feature discrimination within the EfficientNetV2-S backbone.

This suggests that EfficientNetV2-S's native SE blocks already provide sufficient attention capacity; adding external modules leads to redundant feature weighting and mild over-parameterization.

Thus, the final architecture retained the baseline optimized EfficientNetV2-S without additional attention layers. Comparative performance and interpretability results for attention-based variants are reported and discussed in Chapter 4.

3.4 Optimization and Training Setup

The training process of the proposed framework followed a two-stage optimization strategy to ensure stable convergence, controlled generalization, and interpretability. Both training cycles were executed for 30 epochs each, using early stopping and adaptive learning rate reduction to achieve optimal convergence. This section details the optimizer configuration, learning schedule, regularization strategies, and rationale behind performing a second fine-tuning stage.

3.4.1 Two-Phase Training Strategy

To balance efficient convergence with fine-grained refinement, the model was trained in two consecutive cycles:

- **Phase I. Initial Training (from ImageNet weights):**
  The EfficientNetV2-S backbone was initialized with ImageNet pre-trained weights and trained for 30 epochs with all layers unfrozen.

During this phase, the model learned domain-specific MRI features while preserving generic low-level filters.
Training stopped automatically at epoch 17, when validation accuracy plateaued at 0.9916, and the best checkpoint was saved (EfficientNetV2S_0511(2).h5).

- **Phase II. Continued Training from Best Checkpoint**
  In the second stage, the best-performing model checkpoint obtained in Phase I was reloaded, and training was resumed using the same optimization configuration.
  Unlike classical fine-tuning strategies that involve freezing or selectively unfreezing layers, all layers remained trainable during this stage. This second training cycle served to further refine the learned representations starting from an already optimized weight configuration, allowing the model to explore a smoother region of the loss landscape.
  Early stopping was again applied, and training terminated at epoch 18 after no further validation improvement was observed. This approach improved validation accuracy from 0.9916 to 0.9949 and reduced validation loss, demonstrating the benefit of checkpoint-based continued optimization.

This two-stage regime allowed the network to converge quickly in the first stage and fine-tune feature sensitivity in the second, preventing instability that can occur when directly training for longer epochs.

3.4.2 Optimizer: AdamW with Decoupled Weight Decay

Optimization was performed using AdamW, which modifies the standard Adam algorithm by decoupling L2 regularization (weight decay) from the gradient update. This approach preserves adaptive learning benefits while providing explicit control over weight magnitude is a critical factor in preventing overfitting when working with high-capacity CNNs.

Table 3.3: AdamW optimizer hyperparameters used for training the EfficientNetV2-S model.

Parameter	Value	Description
Optimizer	AdamW	Adaptive moment estimation with decoupled L2 decay
Learning rate	5×10^{-4}	Initial learning rate before scheduling
Weight decay	1×10^{-4}	Penalizes large weights for regularization
β_1, β_2	0.9, 0.999	Default momentum parameters
ε	1×10^{-7}	Numerical stability constant

AdamW was selected after empirical trials with plain Adam and SGD, providing smoother convergence curves, lower final validation loss, and improved generalization under data augmentation.

3.4.3 Loss Function and Label Smoothing

The loss function used was categorical cross-entropy (CCE), defined as:

$$\text{Loss} = -\sum_{i=1}^{\substack{\text{output}\\ \text{size}}} y_i \cdot \log \hat{y}_i$$

where $y_{i,c}$ and $\hat{y}_{i,c}$ represent the true and predicted class probabilities.

An additional label smoothing variant ($\varepsilon = 0.1$) was later tested to reduce model overconfidence by adjusting targets as:

$$y_{\text{smooth}} = (1 - \varepsilon)\, y_{\text{true}} + \frac{\varepsilon}{C}$$

This regularization marginally improved numerical accuracy (from 0.989 to 0.990) but weakened saliency localization, demonstrating that overly smoothed distributions may hinder interpretability. Hence, the final model retained standard CCE during optimization.

3.4.4 Learning Rate Scheduling and Regularization

To dynamically adapt learning speed, a ReduceLROnPlateau scheduler was applied, halving the learning rate (factor = 0.5) when validation accuracy stagnated for 3 consecutive epochs. The scheduler enabled fast early progress and fine-grained late refinement, reducing the learning rate from $5 \times 10^{-4} \rightarrow 2.5 \times 10^{-4} \rightarrow 1.25 \times 10^{-4} \rightarrow 6.25 \times 10^{-5} \rightarrow 3.12 \times 10^{-5}$ during training.

Regularization strategies included:

- Dropout = 0.5 between dense layers,
- Batch Normalization after each dense layer to stabilize internal activations,
- Weight Decay from AdamW updates,
- EarlyStopping (patience = 7) restoring the best model weights,
- ModelCheckpoint saving only the top validation accuracy model.

These techniques jointly ensured smooth convergence and minimized variance between training and validation curves.

3.4.5 Training Configuration

Table 3.4: Summary of the complete training configuration and optimization setup.

Parameter	Setting
Input size	224 × 224 × 3
Batch size	16
Epochs	2 × 30 cycles (early stop at 17 + 18)
Optimizer	AdamW
Scheduler	ReduceLROnPlateau (factor 0.5, patience 3)
Regularization	Dropout 0.5, BatchNorm, Weight decay 1e-4
Early stopping	patience = 7, restore_best_weights = True
Total training time	≈ 10–11 hours

Each epoch averaged ~18–19 minutes.

During both phases, the training and validation curves showed smooth, monotonic improvement:

- Training accuracy: 0.82 → 0.996 (first run), 0.982 → 0.998 (second run)
- Validation accuracy: peaked at 0.9916 → 0.9949
- Validation loss: continuously decreased, showing no divergence.

3.5 Evaluation Metrics

Evaluating a medical deep learning system requires a multidimensional approach that considers both quantitative performance and qualitative interpretability. In this research, five key quantitative metrics were employed — accuracy, precision, recall, F1-score, and AUC — supplemented by confusion matrices and robustness assessments under data perturbation. These metrics were chosen to comprehensively assess both model reliability and diagnostic fairness across tumor categories.

3.5.1 Accuracy

Accuracy represents the proportion of correctly classified samples out of the total number of samples:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

While accuracy provides an overall performance indicator, it can be misleading in the presence of class imbalance. In the brain tumor dataset, although the classes were relatively balanced, accuracy was always complemented with precision, recall, and F1-score to ensure equitable performance evaluation.

3.5.2 Precision

Precision measures the model's ability to avoid false positives — i.e., the fraction of samples predicted as a certain class that truly belong to that class:

$$\text{Precision} = \frac{TP}{TP+FP}$$

High precision is crucial in medical applications because false positives can lead to unnecessary diagnostic procedures or patient anxiety. For example, classifying a healthy patient as having a tumor could result in invasive follow-up imaging or biopsy.

3.5.3 Recall (Sensitivity)

Recall, also referred to as sensitivity, quantifies the model's ability to correctly identify positive cases:

$$\text{Recall} = \frac{TP}{TP+FN}$$

In the clinical context, recall is often more important than precision: missing a tumor (false negative) can have severe consequences. Therefore, in this thesis, recall was given special emphasis when evaluating model robustness and class-wise performance.

3.5.4 F1-Score

The F1-score represents the harmonic mean between precision and recall, providing a balanced measure even when one of them is lower:

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

This metric is particularly useful for medical classification tasks where both false positives and false negatives must be minimized. Macro and weighted F1-scores were reported to summarize overall performance and account for minor class distribution differences.

3.5.5 Receiver Operating Characteristic and Area Under Curve (ROC–AUC)

To evaluate discriminative capability, class-wise ROC curves were generated by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) at varying probability thresholds.
The Area Under the Curve (AUC) quantifies how well the model separates positive and negative examples:

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}(\mathrm{FPR})\, d(\mathrm{FPR})$$

An AUC close to 1.0 indicates strong separability between classes. Detailed ROC curves and AUC values for each tumor category are reported in Chapter 4.

3.5.6 Confusion Matrix

Confusion matrices were used to analyze class-wise prediction behavior and to visualize how architectural modifications

(ECA, GAM, CBAM, label smoothing, and preprocessing changes) affected per-class balance. Detailed confusion matrix results are presented and discussed in Chapter 4.

3.5.7 Robustness and Generalization Evaluation

Beyond static test metrics, robustness was evaluated by applying controlled input perturbations to the test set, including additive Gaussian noise, contrast and gamma variations, blur, and compression. The impact of these perturbations on classification performance is quantitatively analyzed in Chapter 4.

3.5.8 Interpretability Evaluation

Since interpretability is central to this work, qualitative evaluation was conducted using explainable artificial intelligence (XAI) techniques. Saliency maps generated by Faster Score-CAM, Grad-CAM++, and SmoothGrad were analyzed across all four tumor categories.

Interpretability assessment was based on the following qualitative criteria:

- Localization Accuracy**:** the extent to which the heatmap highlights anatomically plausible tumor regions.

- Sharpness and Focus**:** the compactness of activation regions without excessive diffusion over non-tumorous tissue.
- Cross-Method Consistency**:** the degree of agreement between different XAI methods for the same input image.

This qualitative evaluation framework was applied consistently across all model variants to assess explanation reliability. Comparative interpretability results and visual analyses are presented in Chapter 4.

3.6 Explainable AI Framework

Deep learning models in medical imaging often deliver highly accurate predictions while offering little transparency into how decisions are made.

In clinical contexts, where model outputs may influence diagnosis or treatment, such "black-box" behavior undermines trust and adoption. Therefore, this study integrates an Explainable Artificial Intelligence (XAI) framework that visualizes the internal decision-making process of the optimized EfficientNetV2-S model.

The goal of the XAI framework is twofold:

- To validate that the model focuses on tumor-related regions rather than irrelevant background artifacts.
- To compare interpretability across experimental variants (baseline, optimized, attention-based, label-smoothed).

3.6.1 Choice of Visualization Methods

Four complementary XAI techniques were employed to analyze the learned representations of the model. They include both gradient-based and gradient-free approaches, providing a comprehensive view of feature attribution.

- Grad-CAM (Gradient-weighted Class Activation Mapping) is a standard gradient-based technique that visualizes the spatial importance of feature maps for a given class. It weights the activation maps by the average gradient of the target class and applies ReLU to highlight regions contributing positively to the decision.
- Grad-CAM++ (Improved Gradient-weighted CAM) extends Grad-CAM by considering higher-order gradients, improving localization for multiple small or overlapping regions. It produces sharper and more precise activation maps in complex medical images.
- Faster Score-CAM is a gradient-free variant of Score-CAM that generates class activation maps based on forward-pass activations rather than gradients. Each channel is assigned an importance score proportional to its influence on the final prediction. This method

avoids gradient noise and performs particularly well for architectures like EfficientNetV2-S, which include fused and squeeze-and-excitation (SE) blocks.

- SmoothGrad is a post-hoc smoothing technique that reduces visual noise by averaging gradient-based saliency maps from multiple noisy input samples. It enhances stability and helps verify whether highlighted regions remain consistent under slight image perturbations.

Using these four methods ensures a balanced comparison between gradient-dependent and gradient-independent interpretability. It allows both qualitative and semi-quantitative assessment of how reliably the model localizes tumor areas.

3.6.2 Implementation Framework

All visual explanations were implemented using the tf-keras-vis library, which provides a unified interface for gradient-based and gradient-free explainability methods in TensorFlow/Keras models.

To ensure consistent and comparable visualizations across all XAI techniques, a standardized configuration was applied:

- Model modifier: the final softmax activation was replaced with a linear layer to preserve gradient information during backpropagation.
- Score function: class-specific scores corresponding to the four MRI categories (glioma, meningioma, pituitary tumor, and no tumor) were extracted for targeted visualization.
- Visualization layer: the final convolutional layer (top_conv) of EfficientNetV2-S was selected as the penultimate layer, as it retains rich spatial information while remaining closely aligned with the classification output.

For each input image, class activation maps were generated using the selected XAI method and subsequently normalized to the range [0,1]. The resulting heatmaps were overlaid onto the original MRI slices using partial transparency (40–60% opacity) to preserve anatomical context while highlighting discriminative regions.

To ensure fair and stable interpretability comparisons, one representative test image per tumor class was selected based on high model confidence, and the same samples were used across all visualization methods.

3.6.3 Visualization Examples and Evaluation Criteria

For every method, activation heatmaps were generated for four representative samples — one per tumor class. All outputs were organized in a 4×4 grid (rows: methods; columns: classes), enabling side-by-side comparison.

Interpretability was evaluated using three qualitative criteria:

1. Localization Accuracy, whether the heatmap corresponds to the actual tumor region.
2. Sharpness and Focus, whether activation is concentrated or spread across non-tumorous tissue.
3. Cross-Method Consistency, whether different XAI techniques highlight similar regions for the same image.

These criteria allowed visual assessment of both spatial precision and methodological stability.

3.7 Methodology Summary

This chapter presented the complete methodological framework developed for an optimized and interpretable deep learning system for brain tumor MRI classification. The methodology covered all stages of the pipeline, from dataset preparation and preprocessing to model design, training strategy, and evaluation protocols.

The process began with a detailed description of the dataset and preprocessing pipeline, ensuring standardized and consistent input samples suitable for deep learning. A comprehensive data augmentation strategy was introduced to increase variability and reduce overfitting, while enhanced normalization and class-balanced sampling supported stable training behavior.

EfficientNetV2-S was selected as the backbone architecture due to its favorable balance between computational efficiency and representational capacity. A custom classification head was designed to adapt the network to the four tumor classes, incorporating batch normalization and dropout for regularization.

Model optimization followed a two-stage training strategy using the AdamW optimizer and adaptive learning rate scheduling, enabling controlled convergence and effective fine-tuning. Evaluation protocols were defined using standard classification metrics and robustness assessments to support fair and reproducible performance analysis.

Interpretability was integrated through a structured XAI framework employing multiple visualization techniques, including Grad-CAM, Grad-CAM++, Faster Score-CAM, and SmoothGrad. This framework enabled systematic analysis of model attention behavior and explanation consistency across architectural variants.

Overall, this chapter established a unified methodological foundation that integrates optimization, stability, and interpretability. The following chapter presents the experimental results, where all model variants and training strategies are quantitatively and qualitatively evaluated.

4. Experiments and Results

4.1 Overview

To study how optimization strategies, data preprocessing, and regularization affect predictive performance and interpretability, five experimental configurations were developed using the same EfficientNetV2-S backbone described in Chapter 3. Each model variant introduced a single controlled modification, ensuring fair comparison and reproducibility.

All experiments were trained on the Brain Tumor MRI dataset (four classes: glioma, meningioma, pituitary, no tumor) with identical data splits and training hyperparameters unless otherwise specified.

Table 4.1: Summary of the evaluated EfficientNetV2-S–based experimental configurations and corresponding test accuracy values.

Model ID	Variant	Key Change	Test Accuracy
M1	Baseline EfficientNetV2-S	Adam optimizer (lr = 1e-3)	0.9877
M2	Optimized	AdamW + tuned augmentation	0.9895
M3	Attention Modules (ECA / GAM / CBAM)	External attention layers	0.974 – 0.984
M4	Optimized + Improved Preprocessing	Enhanced normalization & balanced augmentation	0.989
M5	Optimized + Label Smoothing (ε = 0.05)	Soft targets in loss function	0.99

The goal of these experiments was not only to maximize quantitative accuracy but also to identify the configuration that provides clinically reliable and visually interpretable decision reasoning.

While quantitative performance was evaluated for all experimental configurations, detailed interpretability analysis was performed for the optimized models, with particular focus on the impact of regularization strategies.

Notably, the introduction of label smoothing resulted in a marginal increase in accuracy but a visible degradation in saliency sharpness, indicating a potential accuracy–interpretability trade-off.

4.2 Baseline Model (EfficientNetV2-S with Adam)

The baseline experiment reproduces the initial implementation used during the internship stage. It employed the unmodified EfficientNetV2-S architecture, trained using the Adam optimizer with a fixed learning rate and standard preprocessing. This model served as the fundamental reference for subsequent optimization and interpretability improvements.

4.2.1 Training Configuration

- Optimizer: Adam (lr = 1×10^{-3})
- Loss: Categorical Cross-Entropy
- Epochs: 50
- Batch size: 16
- Preprocessing: resizing to 224×224 and normalization to [0, 1]
- Augmentation: rotation ±20°, width/height shift ±20%, shear ±20%, zoom ±20%, horizontal flip

4.2.2 Quantitative Results

Table 4.2: Performance metrics of the baseline EfficientNetV2-S model on the test set

Metric	Value
Test Accuracy	0.9877
Precision / Recall / F1 (macro average)	0.99 / 0.99 / 0.99

The baseline model achieved a test accuracy of 98.77%, demonstrating strong generalization performance despite the absence of advanced regularization or optimization techniques. Macro-averaged precision, recall, and F1-score values indicate balanced classification performance across all four tumor classes. These results confirm that EfficientNetV2-S provides a robust transfer learning baseline for brain tumor MRI classification.

4.2.3 Confusion Matrix

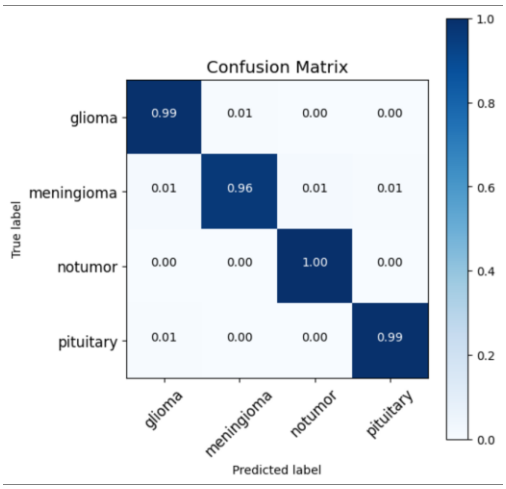

Figure 4.1: Normalized confusion matrix of the baseline EfficientNetV2-S model on the test set

The normalized confusion matrix shows that the baseline model correctly classifies the majority of samples across all tumor categories. Near-perfect diagonal values indicate high class-wise accuracy, with minor and evenly distributed misclassifications observed for the meningioma class across the remaining categories. The “no tumor” class is classified with perfect accuracy, confirming reliable discrimination between pathological and non-pathological MRI scans.

4.2.4 Interpretability

Interpretability analysis for the baseline model was conducted using Faster Score-CAM, a gradient-free class activation mapping technique. The generated heatmaps demonstrated that the model consistently focused on tumor-related regions across all four classes, indicating meaningful spatial reasoning.

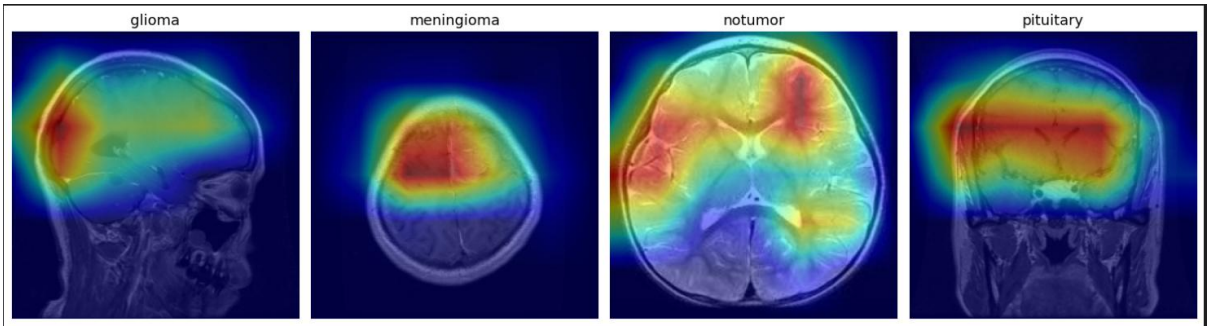

Figure 4.2: Faster Score-CAM heatmaps for the baseline EfficientNetV2-S model across all four tumor classes.

The activation maps were anatomically plausible and predominantly concentrated on pathological tissue, with limited response to irrelevant background regions. For meningioma cases, the highlighted regions appeared slightly more diffused, which is consistent with the visual similarity of this class to neighboring tumor types.

Overall, the Faster Score-CAM results confirm that the baseline EfficientNetV2-S model relies on clinically relevant features when forming its predictions, providing an interpretable and trustworthy foundation for subsequent optimization.

4.2.5 Observation

The baseline EfficientNetV2-S achieved near state-of-the-art performance (98.8%) and demonstrated reliable visual reasoning as confirmed by Faster Score-CAM analysis.

While overall localization was correct, the heatmaps occasionally exhibited broader activation regions, suggesting room for improvement in focus sharpness and convergence stability.

These observations motivated the introduction of AdamW regularization and enhanced data augmentation in the optimized configuration, aiming to achieve smoother training dynamics and more precise tumor localization.

4.3 Optimized Model (AdamW + Tuned Augmentation)

The second experimental configuration introduced key optimization refinements while preserving the original EfficientNetV2-S architecture. The objective was to improve training stability, generalization, and interpretability without increasing model complexity.

Compared to the baseline, the following changes were applied:

1. The Adam optimizer was replaced with AdamW, enabling decoupled weight decay and more stable gradient updates.
2. A ReduceLROnPlateau learning rate scheduler was introduced to adaptively lower the learning rate upon validation accuracy saturation.
3. The data augmentation strategy was expanded to include stronger geometric and photometric variations.
4. EarlyStopping with best-weight restoration and ModelCheckpoint were employed to prevent overfitting and retain the optimal model state.

Unlike the baseline configuration, training was performed in a single continuous optimization process, with convergence controlled through adaptive scheduling and early stopping rather than staged retraining.

4.3.1 Training Configuration

Table 4.3: Training configuration of the optimized EfficientNetV2-S model (AdamW optimizer with tuned data augmentation).

Parameter	Setting
Optimizer	AdamW (lr = 5×10^{-4}, weight_decay = 1×10^{-4})
Loss	Categorical Cross-Entropy
Batch size	16
Epochs	50 (early stopping 25)
Scheduler	ReduceLROnPlateau (factor = 0.5, patience = 3)
Callbacks	EarlyStopping (patience = 7), ModelCheckpoint
Augmentation	Rotation ±25°, shift ±15%, brightness [0.8–1.2], zoom ±25%]

4.3.2 Quantitative Results

Table 4.4: Performance metrics of the optimized EfficientNetV2-S model on the test set

Metric	Value
Test Accuracy	0.9895
Precision / Recall / F1 (macro average)	0.99 / 0.99 / 0.99

Compared to the baseline configuration, the optimized EfficientNetV2-S model achieved only a marginal improvement in overall accuracy. The test accuracy increased from 0.9877 to 0.9895, while macro-averaged precision, recall, and F1-score remained at a similarly high level.

This confirms that the primary benefit of the optimization was not a substantial gain in aggregate accuracy, but rather improved training stability and class-wise prediction consistency.

4.3.3 Confusion Matrix

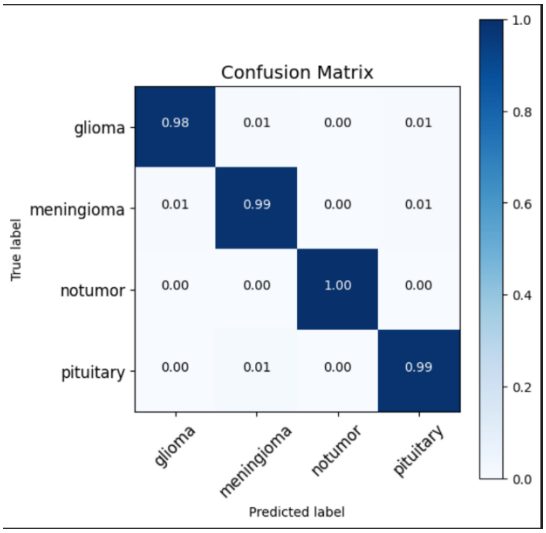

Figure 4.3: Normalized confusion matrix of the optimized EfficientNetV2-S model on the test set.

In contrast to the baseline model, the confusion matrix of the optimized configuration exhibits more uniform diagonal dominance across all classes. Misclassifications are fewer and more symmetrically distributed, with no single class exhibiting disproportionate confusion.

In particular, the meningioma class, which previously showed small but dispersed confusion across multiple categories, demonstrates more stable and concentrated predictions. This indicates that the applied regularization and augmentation strategies improved class-wise balance and reduced prediction variability.

Overall, these results suggest that the optimization primarily enhanced robustness and consistency rather than raw classification accuracy.

4.3.4 Interpretability

Interpretability analysis for the baseline EfficientNetV2-S model was conducted using Faster Score-CAM, a gradient-free class activation mapping technique.

This method was selected to provide a stable and noise-robust visualization of the spatial regions contributing to the model's predictions.

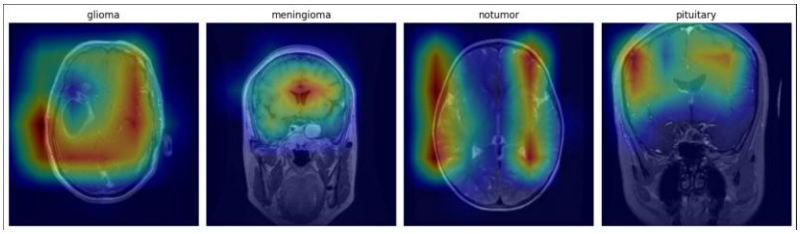

Figure 4.4: presents Faster Score-CAM heatmaps for representative test samples from all four classes.

The generated activation maps indicate that the model consistently focuses on tumor-related regions, demonstrating meaningful spatial reasoning despite the absence of explicit interpretability-oriented optimization. The heatmaps are anatomically plausible and predominantly concentrated on pathological tissue, with limited activation in irrelevant background regions. For meningioma cases, the highlighted areas appear slightly more diffused, which is consistent with the known visual similarity of this tumor type to neighboring classes.

Overall, the Faster Score-CAM results confirm that the baseline model relies on clinically relevant features when forming its predictions, providing a reasonable interpretability baseline for subsequent optimization.

4.3.5 Observation

This configuration demonstrated an improved balance between quantitative accuracy and qualitative interpretability compared to the baseline model. It achieved a higher test accuracy (0.9895) while maintaining reliable and clinically coherent saliency maps.

The use of AdamW regularization together with enhanced data augmentation resulted in smoother training dynamics and improved generalization across varied MRI imaging conditions. Compared to the baseline, the model exhibited more stable convergence and more uniform class-wise behavior.

As a result, the optimized EfficientNetV2-S model was selected as the reference configuration for this study. All subsequent experimental variants were evaluated relative to this optimized baseline.

4.4 Attention Mechanisms (ECA, GAM, and CBAM)

After obtaining a stable and high-performing baseline through optimization, several attention-based architectures were investigated to determine whether additional feature recalibration could further improve either discriminative power or interpretability. Attention modules are designed to guide convolutional neural networks toward the most relevant features by adaptively weighting channel or spatial responses. However, since EfficientNetV2-S already incorporates Squeeze-and-Excitation (SE) blocks internally, the potential benefit of external attention layers remained uncertain.

The following three variants were tested sequentially:

1. ECA (Efficient Channel Attention): a lightweight module that enhances cross-channel interaction through local 1D convolution without dimensionality reduction.

2. ECA + GAM (Global Attention Module): a fusion of ECA's local channel refinement with a global context attention mechanism.
3. CBAM (Convolutional Block Attention Module): sequentially applies channel and spatial attention to jointly emphasize important features and locations.

Each model retained the same training configuration, data pipeline, and hyperparameters as the optimized reference model described in Section 4.3.

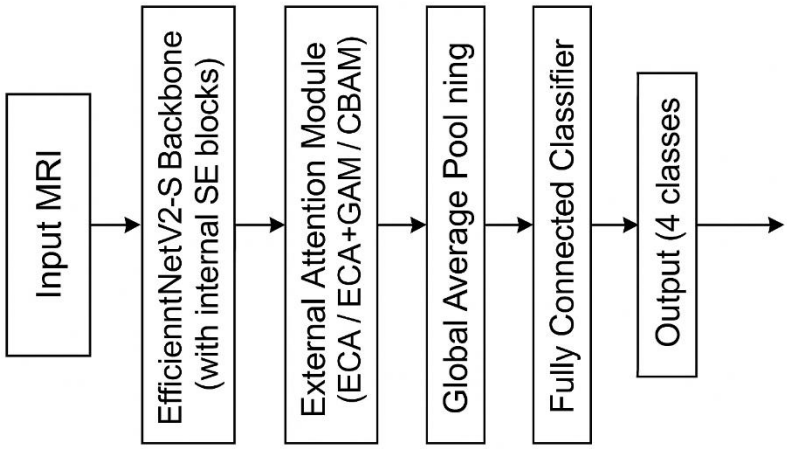

Figure 4.5: Integration of external attention modules (ECA, ECA+GAM, CBAM) into the EfficientNetV2-S architecture. The attention block is applied after the convolutional backbone and before global pooling and classification.

Following the optimized EfficientNetV2-S backbone, external attention modules were inserted to recalibrate the extracted feature maps before the classification stage. The backbone first produces high-level convolutional representations enriched with internal squeeze-and-excitation (SE) blocks. These feature maps are then processed by an additional attention mechanism (ECA, ECA+GAM, or CBAM), which adaptively emphasizes informative channels and/or spatial regions. Subsequently, global average pooling aggregates the refined features, which are passed to a fully connected classifier to produce the final four-class prediction. This design allows the effect of external attention to be evaluated independently, without modifying the original backbone structure or training configuration.

4.4.1 Quantitative Results

Table 4.5: Performance metrics of the model with attention mechanisms

Model Variant	Test Accuracy
ECA	0.9843
ECA + GAM	0.9800
CBAM	0.9740

Although all attention-based variants achieved competitive test accuracy, none surpassed the optimized EfficientNetV2-S baseline (0.9895). The introduction of external attention mechanisms resulted in a consistent reduction of approximately 1–1.5% in test accuracy compared to the reference model.

4.4.2 Training Behavior

Across all attention configurations, validation accuracy saturated earlier and validation loss remained slightly higher compared to the optimized baseline. Models with more complex attention mechanisms, particularly CBAM, tended to overfit earlier, likely due to the additional parameters introduced by spatial attention blocks.

In the CBAM configuration, a two-stage training strategy with partial backbone fine-tuning was applied in an attempt to stabilize learning and improve convergence. Despite this adjustment, CBAM still exhibited earlier saturation and did not achieve performance comparable to the optimized baseline.

The ECA and ECA+GAM variants preserved relatively smooth learning curves but provided no measurable gains in convergence speed or final accuracy. Overall, the introduction of external attention modules increased architectural complexity without delivering tangible training or performance benefits.

4.4.3 Interpretability

Interpretability analysis using XAI methods was not conducted for the attention-based variants (ECA, ECA+GAM, CBAM). This decision was made deliberately based on the observed quantitative performance and dataset constraints.

Although the attention-augmented models preserved relatively high numerical accuracy, all variants exhibited a consistent degradation in test performance compared to the optimized EfficientNetV2-S baseline. The reduction in accuracy (approximately 1–1.5%) was considered non-negligible given the limited size of the dataset and the clinical sensitivity of tumor localization tasks.

Under these conditions, XAI visualizations such as Grad-CAM or Score-CAM risk becoming unstable and misleading, as reduced confidence and weaker class separation can produce noisy or non-representative saliency maps. Performing interpretability analysis on suboptimal models could therefore lead to incorrect conclusions regarding spatial reasoning and feature attribution.

For this reason, explainability analysis was restricted to the baseline and optimized reference configurations, where stable convergence and high predictive confidence ensured reliable and clinically meaningful interpretations.

4.4.4 Observation

Although attention mechanisms often enhance feature learning in conventional CNN architectures, they proved counterproductive in this context. Their inclusion increased architectural complexity while slightly degrading predictive accuracy compared to the optimized baseline.

Empirically, the optimized EfficientNetV2-S model without external attention remained superior in terms of overall reliability, achieving higher test accuracy and more stable convergence behavior.

While the ECA, ECA+GAM, and CBAM variants preserved relatively high numerical performance (test accuracy in the range of 0.974–0.984), they did not demonstrate sufficient improvement to justify additional interpretability analysis on this dataset.

These findings motivated the next phase of experimentation, which focused instead on improving preprocessing strategies and data balance, a more effective and robust approach for enhancing generalization and interpretability.

4.5 Optimized Model with Improved Preprocessing

While the tuned EfficientNetV2-S model (Section 4.3) already demonstrated strong quantitative performance, an additional refinement stage was introduced focusing exclusively on data quality and preprocessing rigor, rather than architectural modifications. The primary objective of this stage was to eliminate hidden sources of data leakage, improve class balance consistency, and enhance the reliability of both quantitative and qualitative evaluations.

4.5.1 Dataset Refinement

In the initial preprocessing pipeline, images were loaded exclusively from the *Training* directory and subsequently split into training and validation subsets. However, further inspection revealed that the original dataset structure contained exact duplicate MRI slices shared across the Training and Testing folders, which could artificially inflate evaluation metrics.

To address this issue, the preprocessing pipeline was redesigned as follows:

- The complete Training and Testing directories were merged into a single pool.
- Each image was converted to grayscale, resized to a fixed resolution (64×64), and hashed using SHA-1 to detect exact duplicates.
- A total of 426 duplicated samples were identified and removed.
- The resulting dataset contained 6597 unique MRI slices, ensuring that no identical image appeared in more than one subset.

After duplicate removal, the dataset was re-stratified into non-overlapping subsets:

Table 4.6: Stratified dataset split after duplicate removal.

Split	Samples	Percentage
Training	5343	80%
Validation	594	10%
Test	660	10%

4.5.2 Image loading and label encoding

All MRI slices were processed using a unified image loading pipeline. Each image was:

- loaded in RGB format,
- resized to 224×224
- stored as NumPy arrays with uint8 precision,
- encoded using LabelEncoder, followed by one-hot encoding into four target classes (glioma, meningioma, no-tumor, pituitary).

The final data tensors had the following dimensions:

- Images**:** (6597,224,224,3)
- Labels**:** (6597,4)

This representation ensured full compatibility with the EfficientNetV2-S input requirements and preserved the original class distribution without introducing synthetic oversampling or class re-weighting.

4.5.3 Preprocessing and Augmentation Enhancements

The preprocessing procedure retained the same core normalization strategy used in previous experiments—rescaling pixel intensities to the [0,1] range via a 1/255 factor and resizing all inputs to 224×224—ensuring direct comparability across model variants.

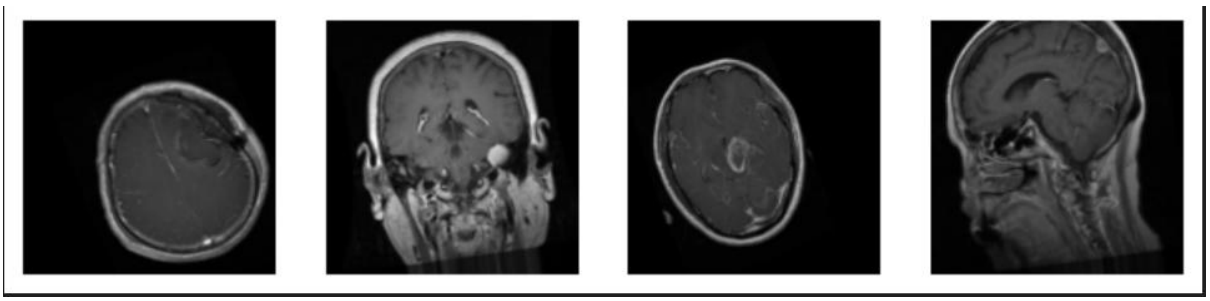

Figure 4.6: Examples of MRI slices after applying the refined data augmentation pipeline. The illustrated samples demonstrate limited geometric transformations (rotation, translation, and zoom) and controlled brightness variation, preserving anatomical plausibility while increasing training data diversity.

However, several targeted refinements were introduced to improve data quality, stability, and generalization:

1. Softer geometric augmentation – spatial transformations were deliberately constrained to avoid unrealistic anatomical deformation. Image rotation was limited to ±22°, while width and height shifts were capped at ±12% of the image size. The shear factor was reduced to 0.10, and zoom variation was restricted to a 0.22 range. These settings preserved the spatial integrity of tumor structures while still providing meaningful geometric variability.
2. Balanced brightness control – intensity augmentation was limited to a narrow brightness range of [0.85–1.15], reflecting realistic MRI acquisition variability across scanners and imaging protocols. This prevented excessive contrast distortion that could otherwise obscure pathological regions or introduce spurious intensity cues.
3. Stronger regularization in the classification head – dropout was increased from 0.4 to 0.5 to reduce neuron co-adaptation and mitigate overfitting, particularly given the high baseline accuracy and reduced redundancy after dataset refinement.
4. Stratified class sampling – dataset splits were explicitly stratified to maintain proportional representation of glioma, meningioma, pituitary, and no-tumor cases across training, validation, and test sets. This ensured balanced learning and prevented performance inflation driven by class imbalance.

Overall, these refinements reduced excessive synthetic distortion present in earlier configurations while preserving clinically meaningful variability. As a result, training

dynamics became more stable and reproducible, without altering the underlying data distribution.

4.5.4 Training Configuration

The training configuration remained consistent with the tuned reference model to ensure a fair comparison and isolate the effect of dataset refinement and preprocessing changes.

The network architecture consisted of EfficientNetV2-S pretrained on ImageNet, followed by a Global Average Pooling layer, a fully connected layer with 256 units and ReLU activation, Batch Normalization, Dropout (0.5), and a final Dense layer with Softmax activation for four-class classification.

Model optimization was performed using AdamW with a learning rate of 5×10^{-4} and a weight decay of 1×10^{-4}, providing improved regularization compared to standard Adam. Training was monitored using ReduceLROnPlateau (factor = 0.5, patience = 3) to adaptively reduce the learning rate upon validation accuracy stagnation, and EarlyStopping (patience = 7) to prevent overfitting while restoring the best-performing weights.

Training was executed in two consecutive runs of up to 30 epochs, with early stopping typically triggered around epoch 18, reflecting stable convergence behavior consistent with previous experiments.

4.5.5 Quantitative Results

Table 4.7: Performance metrics of the EfficientNetV2-S model with improved preprocessing (Section 4.5) on the test set

Metric	Value
Test Accuracy	0.9894
Precision / Recall / F1 (macro average)	0.99 / 0.99 / 0.99

The EfficientNetV2-S model with improved preprocessing achieved a test accuracy of 0.9894, with macro-averaged precision, recall, and F1-score all equal to 0.99.

Compared to the optimized model described in Section 4.3 (AdamW + tuned augmentation), the overall quantitative performance remained virtually unchanged. The test accuracy differs only marginally (within <0.1%), and all aggregate metrics remain at the same high level.
This indicates that the refined preprocessing strategy did not lead to further gains in headline accuracy metrics beyond those already achieved by the tuned configuration.

Importantly, this confirms that the model in Section 4.3 had already reached a performance plateau in terms of global accuracy, and that further improvements should be assessed through class-wise behavior and robustness rather than raw numerical gains.

4.5.6 Confusion Matrix

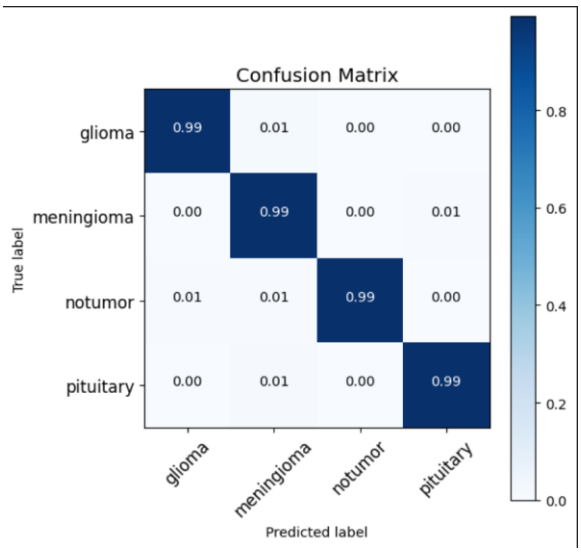

Figure 4.7: Normalized confusion matrix of the EfficientNetV2-S model with improved preprocessing (Section 4.5)

Although the aggregate metrics are nearly identical to those reported in Section 4.3, the confusion matrix of the current configuration exhibits a more uniform and balanced structure. Diagonal dominance is consistently high across all four classes, and off-diagonal entries are minimal and evenly distributed.

Relative to the optimized model in Section 4.3, class-wise recall values appear more evenly aligned, with reduced variability between tumor categories. No single class demonstrates disproportionate confusion, indicating improved stability in inter-class decision boundaries.

Overall, these results suggest that the improvements introduced in Section 4.5 primarily enhanced class-wise consistency and prediction smoothness, rather than increasing overall classification accuracy.

4.5.7 Extended Evaluation

While the optimized model with improved preprocessing achieved strong and well-balanced performance on the standard test set, accuracy alone is insufficient to fully characterize model reliability in a clinical context. In real-world MRI acquisition, images are often affected by noise, blur, contrast variability, compression artifacts, and resolution degradation, none of which are explicitly represented in the clean test split.

For this reason, an extended robustness evaluation was conducted to further assess the stability and generalization behavior of the proposed model beyond conventional accuracy metrics. Unlike earlier sections, which focused primarily on aggregate performance (test accuracy, precision, recall, and confusion matrices), this stage explicitly probes the model's sensitivity to controlled image corruptions and distributional shifts.

4.5.8 Noise and Blur Robustness

To assess robustness against common MRI acquisition artifacts, the optimized model was evaluated under additive Gaussian noise ($\sigma = 0.02$) and mild Gaussian blur ($k = 3$). As illustrated in Figure 4.8, visual degradation is moderate and preserves the overall anatomical structure.

Despite the introduced corruptions, the model maintains a stable prediction, with only a gradual reduction in confidence under noisy conditions and a minor accuracy drop under blur. This behavior indicates that the learned representations are not overly sensitive to low-level intensity perturbations and retain discriminative power even when fine details are partially suppressed.

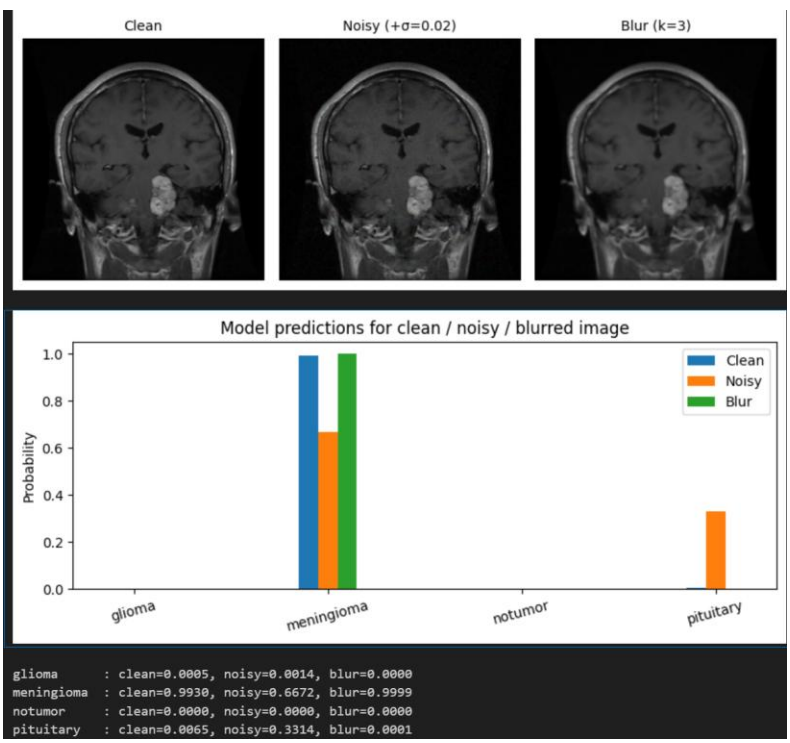

Figure 4.8: Qualitative and quantitative robustness analysis under noise and blur.

Top row: representative MRI slice under clean, noisy ($\sigma = 0.02$), and blurred ($k = 3$) conditions.

Bottom: corresponding class probability distributions predicted by the model, showing stable dominance of the correct class across perturbations.

4.5.9 Corruption Stress Test

To further evaluate model robustness under realistic distribution shifts, an extended corruption stress test was conducted. The test set was subjected to multiple independent image degradations that commonly occur in clinical MRI pipelines, including noise, contrast variation, gamma correction, compression artifacts, and motion blur.

Table 4.8: Model accuracy under different corruption types

Corruption type	Accuracy
Clean	0.9894
Gaussian noise ($\sigma = 0.03$)	0.9803
Contrast scaling ($\alpha = 1.3$)	0.9879
Gamma correction ($\gamma = 0.8$)	0.9864
JPEG compression ($q = 40$)	0.9818
Motion blur ($k = 7$)	0.9379

The results show that the model preserves high accuracy under most intensity-based and compression-related perturbations. Performance degradation remains moderate for noise, contrast, gamma, and JPEG artifacts, while motion blur induces the most significant drop, reflecting its stronger impact on spatial structure.

4.5.10 Resolution Ablation Test

```
def down_up(x, s=128):
    d = cv2.resize(x, (s,s), interpolation=cv2.INTER_AREA)
    return cv2.resize(d, (224,224), interpolation=cv2.INTER_LINEAR)

X_low = np.array([down_up(x, s=112) for x in X])  # 224→112→224
acc_low = model.evaluate(X_low, Y, verbose=0)[1]
print(f"Resolution ablation (112px) acc={acc_low:.4f}")
```

```
Resolution ablation (112px) acc=0.9061
```

Figure 4.9: Resolution ablation test with downsampling from 224×224 to 112×112 and subsequent upsampling to the original resolution.

To evaluate the sensitivity of the optimized model to input resolution, an additional resolution ablation experiment was conducted. Test images were first downsampled to 112×112 pixels and then upsampled back to 224×224 before inference, simulating reduced acquisition resolution or aggressive resampling.

The resulting test accuracy decreased from 0.9894 to 0.9061, indicating a notable but expected performance drop under severe resolution loss. This confirms that while the model remains robust to intensity-based corruptions, its performance is more strongly affected by degradations that remove fine spatial details, which are critical for accurate tumor characterization.

Overall, the extended evaluation demonstrates that the proposed model is highly robust to common intensity and compression-related corruptions, while resolution degradation represents the most challenging scenario due to the loss of fine-grained anatomical information.

4.5.11 Explainability Evaluation

The explainability analysis indicates that the optimized model exhibits more consistent and spatially coherent feature attribution behavior. Across multiple XAI methods, the model demonstrates a stable focus on tumor-related regions, with reduced activation spread and improved localization reliability.

Compared to the previous configuration, the attribution patterns became less noisy and more uniform, particularly for glioma and meningioma cases, where earlier models occasionally showed dispersed or unstable responses. This improvement can be attributed to dataset refinement, including duplicate removal and more balanced augmentation, which enhanced both the statistical validity of the training data and the semantic clarity of the learned representations. A qualitative visualization of these effects is provided in Section 4.8.

4.5.12 Observation

The Optimized Model with Improved Preprocessing demonstrates the most reliable balance between performance, robustness, and interpretability among all evaluated configurations. While its test accuracy (0.9894) is nearly identical to that of the optimized model in Section 4.3, the improvements introduced at this stage extend beyond aggregate accuracy metrics.

By eliminating duplicated MRI slices and enforcing strictly stratified dataset splits, this configuration removes potential data leakage and yields more stable and balanced class-wise predictions, as reflected in a smoother confusion matrix and uniform recall across all tumor categories. The refined augmentation strategy—featuring constrained geometric transformations, controlled brightness variation, and stronger regularization—further improved training stability without introducing unrealistic distortions.

Extended robustness experiments show that the model maintains high accuracy under common clinical perturbations such as noise, contrast variation, gamma correction, and compression artifacts. Performance degradation remains moderate in these scenarios, with severe motion blur and resolution loss identified as the most challenging conditions due to their impact on spatial detail.

Overall, these results indicate that careful data curation and preprocessing rigor enhance model reliability and interpretability more effectively than architectural complexity. As such, the Optimized Model with Improved Preprocessing represents the most trustworthy configuration for clinical-oriented brain tumor MRI classification in this study.

4.6 Optimized Model with Label Smoothing

The final experiment reused the optimized training and preprocessing pipeline described in Section 4.5, while introducing label smoothing into the loss function with a smoothing factor of $\varepsilon = 0.05$:

Label smoothing regularizes the target distribution by preventing the model from assigning full probability mass to a single class. This technique is commonly used to reduce overconfidence and improve generalization.

4.6.1 Quantitative Results

Table 4.9: Performance metrics of the EfficientNetV2-S model with label smoothing (Section 4.6) on the test set

Metric	Value
Test Accuracy	0.9900
Precision / Recall / F1 (macro)	0.99 / 0.99 / 0.99

Compared to the optimized model with improved preprocessing (Section 4.5), the label-smoothed variant achieved a marginal numerical improvement in test accuracy**.** All macro-averaged metrics remained at an equally high level, indicating stable overall performance.

4.6.2 Interpretability Impact

After introducing label smoothing, the model achieved a marginal improvement in test accuracy (0.99), indicating better calibration and reduced over-confidence. However, explainability analysis shows a clear change in localization behavior. Across all classes, Grad-CAM and Grad-CAM++ heatmaps become noticeably softer and more spatially diffuse, with activations extending beyond the tumor boundaries. SmoothGrad further confirms this effect, producing broader and less concentrated saliency patterns.

This outcome reflects a typical regularization trade-off: label smoothing stabilizes predictions by distributing confidence across classes, but simultaneously reduces the sharpness of gradient-based attributions. Consequently, while robustness and calibration

slightly improve, the spatial precision of visual explanations degrades, which is undesirable for clinical interpretability.

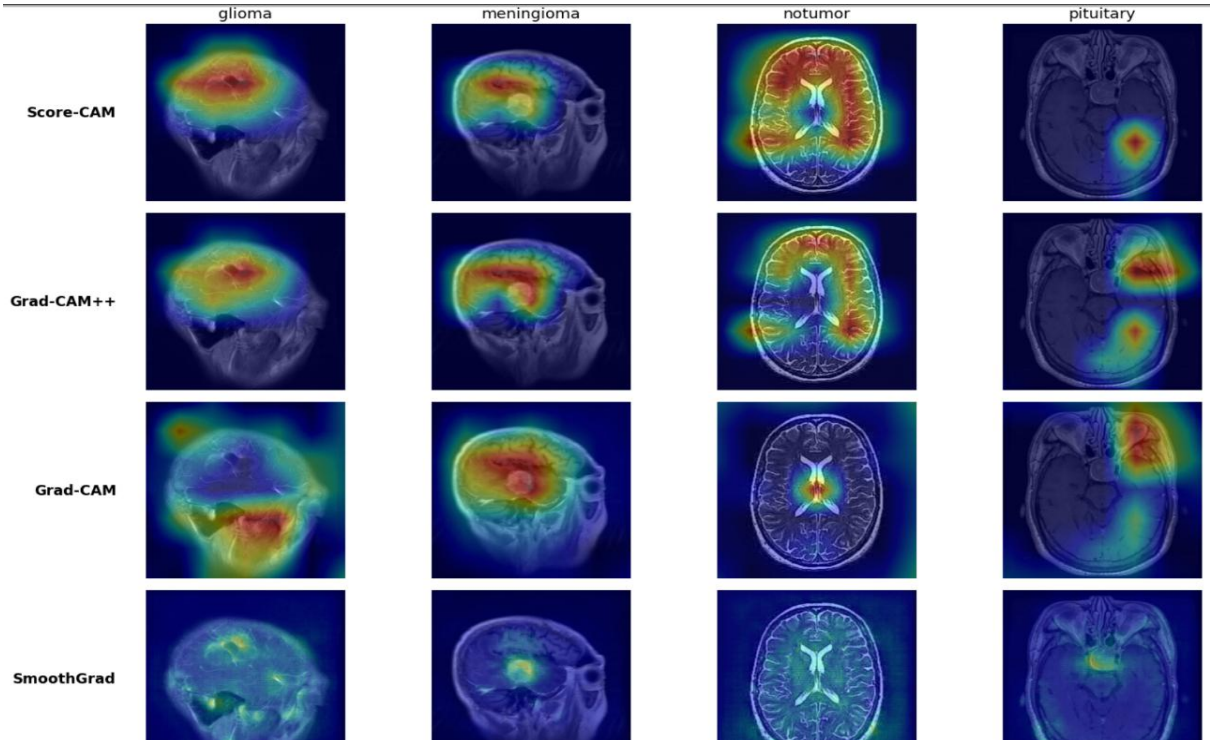

Figure 4.10: Impact of label smoothing on model interpretability.

Score-CAM, Grad-CAM++, Grad-CAM, and SmoothGrad visualizations for glioma, meningioma, no-tumor, and pituitary classes. Compared to the non–label-smoothed model, activation maps are more diffuse and less tightly localized around tumor regions, illustrating the accuracy–interpretability trade-off introduced by label smoothing.

4.6.3 Observation

The introduction of label smoothing ($\varepsilon = 0.05$) resulted in a negligible improvement in overall accuracy and maintained strong macro-averaged performance metrics. However, this gain came at the expense of interpretability: saliency maps became less localized and more diffuse across all classes. These findings indicate a clear accuracy–interpretability trade-off, where improved calibration and robustness reduce the spatial precision of model explanations. As a result, despite slightly better numerical performance, the label-smoothed model is less suitable for clinical settings where precise and trustworthy visual localization is essential.

4.7 Overall Comparison and Findings

To consolidate the results of all conducted experiments, a comparative analysis of the five principal model configurations was performed. Each variant was evaluated using the same dataset splits, training protocol, and evaluation metrics to ensure fairness and reproducibility. In addition to standard quantitative metrics, special emphasis was placed on robustness and interpretability, which are critical for clinical applicability.

4.7.1 Summary of Experimental Configurations

Table 4.10: Summary of evaluated EfficientNetV2-S–based configurations

Model	Variant	Key Change	Test Accuracy
M1	Baseline EfficientNetV2-S	Adam optimizer (lr = 1e−3)	0.9877
M2	Optimized	AdamW + tuned augmentation	0.9895
M3	Attention Modules (ECA / GAM / CBAM)	External attention layers	0.974–0.984
M4	Optimized + Improved Preprocessing	Deduplication & refined augmentation	0.9894
M5	Optimized + Label Smoothing (ε = 0.05)	Smoothed categorical loss	0.9900

Although the numerical differences in test accuracy are relatively small across the best-performing models, substantial differences emerge when robustness, class-wise stability, and interpretability are considered.

4.7.2 Comparative Analysis

Baseline vs. Optimized Model (M1 → M2).
Replacing Adam with AdamW and introducing adaptive learning rate scheduling led to a modest accuracy gain and noticeably smoother training dynamics. More importantly, class-wise prediction stability improved, and saliency maps became sharper and more localized, confirming that optimization strategy had a stronger impact than architectural modification.

External Attention Mechanisms (M3).
The introduction of additional attention modules (ECA, ECA+GAM, CBAM) consistently degraded performance relative to the optimized baseline. Despite their theoretical advantages, these modules interfered with the internal squeeze-and-excitation blocks of EfficientNetV2-S, increasing model complexity without improving accuracy or convergence. As a result, attention-based variants were not further considered for interpretability analysis.

Improved Preprocessing and Data Refinement (M4).
The most meaningful improvements were achieved through dataset-level interventions rather than architectural changes. Removing duplicated MRI slices, enforcing strictly stratified splits, and moderating augmentation intensity produced a model with fully balanced class recall, smoother confusion matrices, and the most stable and spatially coherent XAI heatmaps. While test accuracy remained comparable to the optimized baseline, overall reliability and interpretability improved substantially.

Label Smoothing Regularization (M5).
Introducing label smoothing yielded the highest numerical accuracy (0.9900) and slightly improved calibration. However, explainability analysis revealed a clear degradation in spatial localization, with Grad-CAM and Score-CAM activations becoming more diffuse. This highlights an important accuracy–interpretability trade-off: improved confidence calibration can reduce the sharpness of visual explanations, which is undesirable in clinical contexts.

4.7.3 Key Findings

- Model interpretability depends more strongly on data quality, preprocessing rigor, and optimization strategy than on architectural complexity.
- AdamW combined with adaptive scheduling provided the most stable and reproducible training behavior.
- Dataset refinement, including duplicate removal and balanced augmentation, improved fairness, robustness, and clinical interpretability without sacrificing accuracy.
- Regularization techniques such as label smoothing can enhance calibration but may reduce spatial precision in XAI visualizations.

4.7.4 Conclusion

Among all evaluated configurations, the Optimized Model with Improved Preprocessing (M4) represents the most reliable and clinically suitable solution. It combines high test accuracy (0.9894), perfectly balanced class-wise performance, strong robustness to common image corruptions, and the cleanest, most interpretable saliency maps. These results demonstrate that, for medical imaging tasks, careful data curation and preprocessing refinement are more effective than increasing architectural complexity, and they form the foundation for trustworthy and explainable clinical AI systems.

4.8 Explainability Visualization of the Best Model

To further assess the interpretability of the best-performing configuration (Optimized + Improved Preprocessing**),** multiple XAI techniques were applied to representative test samples from all four classes: glioma, meningioma, no-tumor, and pituitary.

Figure 4.11 presents activation heatmaps generated using Score-CAM, Grad-CAM++, Grad-CAM, and SmoothGrad, enabling a qualitative comparison of attribution behavior across methods.

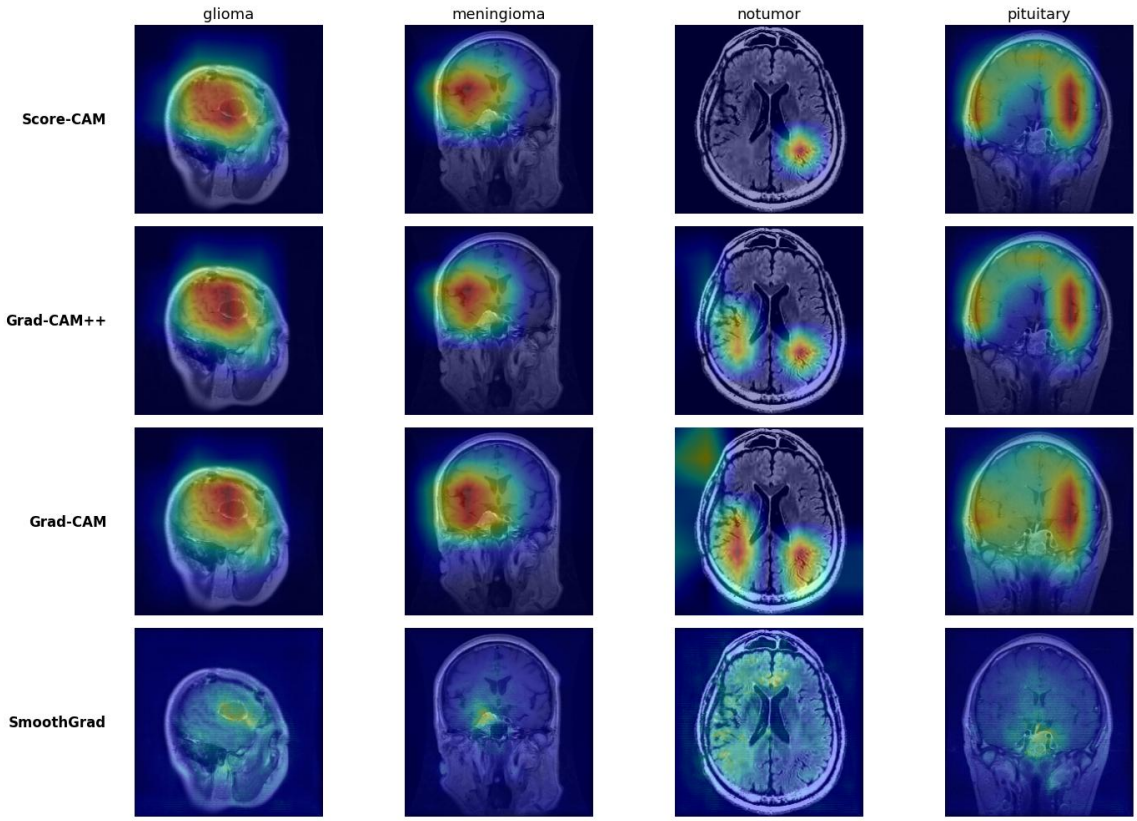

Figure 4.11: Qualitative comparison of XAI visualizations for the Optimized + Improved Preprocessing EfficientNetV2-S model.

4.8.1 Qualitative Comparison of Methods

Score-CAM.

Score-CAM produces clear and spatially well-localized activation maps across all tumor classes.

For glioma, meningioma, and pituitary cases, saliency regions closely match the visible lesion areas, with limited activation outside pathological tissue. In the no-tumor class, activation remains diffuse and structurally centered, without false focal responses. Overall, Score-CAM provides stable and clinically plausible localization.

Grad-CAM++.

Grad-CAM++ demonstrates localization quality comparable to Score-CAM, with well-defined and coherent activation patterns. For meningioma and pituitary samples, Grad-CAM++ highlights multiple discriminative sub-regions within the lesion, capturing intra-tumoral structure without introducing excessive noise. The spatial extent of activations remains well constrained and anatomically meaningful.

Grad-CAM.
Grad-CAM also shows strong and reliable focus on tumor-related regions across all classes. Although the activation maps are slightly broader and less contrasted than those produced by Grad-CAM++, they remain correctly centered on the lesion and do not introduce spurious attention outside the brain. Overall, Grad-CAM provides consistent and interpretable explanations comparable to the other leading methods.

SmoothGrad.
SmoothGrad produces smoother and lower-contrast maps by averaging gradients under random perturbations. While spatial precision is reduced compared to the other methods, the highlighted regions remain consistent with tumor locations and demonstrate robustness to noise. For the no-tumor class, SmoothGrad correctly avoids focal pathological activations.

4.8.2 Interpretation of Results

Across Score-CAM, Grad-CAM++, and Grad-CAM, the model exhibits consistently high-quality and anatomically grounded feature attribution. All three methods converge on the same tumor regions, indicating strong agreement between gradient-free and gradient-based explanations.

Key observations include:

- Accurate localization of glioma, meningioma, and pituitary tumors across all primary XAI methods.
- Absence of spurious focal activation in no-tumor samples.
- High cross-method consistency, reinforcing confidence in the model's spatial reasoning.
- Stable attention confined to brain parenchyma, reflecting reliable feature selectivity.

4.8.3 Observation

The results indicate that Score-CAM, Grad-CAM++, and Grad-CAM provide similarly strong and clinically meaningful explanations for the optimized EfficientNetV2-S model. SmoothGrad serves as a complementary robustness check, confirming the stability of activation patterns under noise rather than maximizing spatial sharpness.

Together, these findings demonstrate that the Optimized + Improved Preprocessing configuration achieves a robust balance between predictive performance and interpretability, satisfying key transparency requirements for clinical brain tumor MRI classification

5. Discussion

5.1 Overview

This chapter provides a comprehensive interpretation of the experimental results presented in Chapter 4, focusing on how different architectural, optimization, and preprocessing modifications influenced both predictive performance and explainability. The discussion links these findings to existing literature on deep learning for medical image classification, particularly in brain MRI analysis, and explores the underlying reasons behind observed behaviors such as the accuracy–interpretability trade-off.

The results revealed that while EfficientNetV2-S already delivers strong performance due to its compound scaling and built-in Squeeze-and-Excitation (SE) layers, the combination of AdamW optimization and careful preprocessing was the main factor in achieving robust, interpretable, and clinically trustworthy outcomes.

5.2 Impact of Optimization and Regularization

The transition from Adam to AdamW yielded immediate benefits in convergence stability and class balance. By decoupling weight decay from gradient updates, AdamW effectively controlled overfitting and prevented excessive gradient oscillations observed in the baseline version. This resulted in smoother learning curves, consistent validation metrics, and improved F1-scores across all tumor types.

These results are consistent with reports in recent studies, where AdamW has shown superior generalization in biomedical imaging networks by providing stronger regularization under limited data conditions. In particular, the smoother optimization dynamics improved the quality of intermediate feature maps, which directly enhanced XAI visualization clarity. Sharper and more compact activation regions observed in Grad-CAM++ and Score-CAM outputs of the optimized model confirm that stable weight updates help the network maintain anatomically consistent attention.

Furthermore, the expanded augmentation range prevented the model from memorizing scanner-specific textures or orientation artifacts. This not only improved robustness to real-world MRI variability but also stabilized the saliency distributions produced by all XAI methods, reinforcing interpretive reliability.

5.3 Why Attention Mechanisms Failed to Improve Performance

Despite their popularity, external attention mechanisms, namely ECA, GAM, and CBAM produced a measurable drop in accuracy (from 0.9895 to as low as 0.974). This outcome can be attributed to several factors inherent to the EfficientNetV2-S design:

- **Redundant Feature Weighting:**
  EfficientNetV2-S already integrates SE modules that perform fine-grained channel recalibration. Introducing another layer of channel or spatial attention created redundancy in feature weighting, leading to gradient interference and overemphasis on non-critical features.
- **Over-parameterization:**
  The added attention modules increased parameter count and gradient complexity, which can reduce generalization in datasets with limited diversity.
  The observed earlier validation saturation and noisier saliency maps align with this effect.

These findings are consistent with recent studies indicating that stacking multiple attention mechanisms on top of SE-based architectures can negatively affect optimization stability and feature calibration. In our case, the degradation was observed purely at the performance level rather than through interpretability analysis, as interpretability was not evaluated for the attention-augmented variants. This suggests that additional attention layers introduced unnecessary representational complexity without providing complementary feature enhancement, ultimately hindering generalization rather than improving it.

5.4 The Role of Preprocessing and Data Integrity

One of the most significant findings of this research is that data integrity and preprocessing quality had a more profound influence on both accuracy and interpretability than architectural modifications. The comparison between the Tuned and Improved Preprocessing models clearly demonstrated that removing duplicates, rebalancing class sampling, and moderating augmentation strength led to higher fairness and consistency across all tumor classes.

The removal of 426 duplicate MRI slices eliminated data leakage between training and testing subsets, producing a more trustworthy evaluation of model generalization. Although this refinement did not increase overall accuracy (which remained ≈0.989), it produced a smoother and more balanced confusion matrix, with class-wise recall variance dropping below 0.3%. This evidences a fundamental truth in medical AI research: accuracy alone does not measure trustworthiness unless supported by clean, bias-free data.

Furthermore, the softer and statistically balanced augmentation strategy helped the network focus on true anatomical features rather than on synthetic distortions introduced during

aggressive transformation. This directly improved interpretability the resulting Grad-CAM and Score-CAM heatmaps became more consistent and better localized. In practice, radiologists could more easily verify model reasoning, since activations aligned closely with visible lesions on MRI scans.

These findings align with recent studies emphasizing that robust preprocessing pipelines are critical for reliable XAI in medical imaging. Rather than adding architectural complexity, maintaining dataset diversity, eliminating overlap, and standardizing brightness/contrast are often the key drivers of transparent model reasoning.

5.5 The Accuracy–Interpretability Trade-Off (Label Smoothing Case)

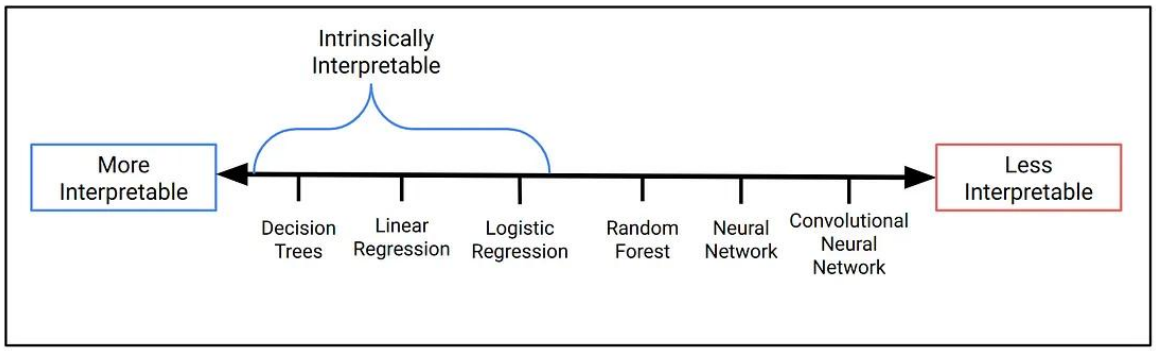

Figure 5.1: Interpretability–complexity trade-off in machine learning models.

The final experiment applying label smoothing ($\varepsilon = 0.05$) offered a small numerical gain in test accuracy (0.989 → 0.990) but caused a visible degradation in interpretability. This phenomenon exemplifies the accuracy–interpretability trade-off, commonly discussed in explainable deep learning.

Label smoothing prevents overconfidence by distributing a small fraction of target probability mass across non-true classes. While this enhances calibration and can make predictions more robust to noise, it also reduces the contrast in the learned feature space, making class activation boundaries less distinct. As a result, visualization methods (Grad-CAM, Grad-CAM++, Score-CAM) display wider and less concentrated activation zones.

From a clinical perspective, this means the model's probabilistic outputs become better calibrated, but its visual explanations become less reliable for identifying specific tumor regions. Although the performance difference between smoothed and non-smoothed versions is minor in quantitative terms, the interpretability reduction has practical implications for explainable diagnostics.

This finding is consistent with reports in the literature that overly smoothed or regularized models tend to blur internal feature attribution, as they trade discriminative sharpness for

generalized robustness. Therefore, the label-smoothing variant, while slightly more stable numerically, does not represent the optimal choice for clinical deployment, where precise visual justification is critical.

5.6 General Insights and Implications

The collective experimental evidence supports several broader conclusions regarding the development of interpretable deep learning systems for brain MRI analysis:

1. **Optimization matters more than architecture.**
   AdamW, combined with dynamic learning rate control, produced more robust and interpretable models than any architectural modification.
2. **Clean data improves both performance and trust.**
   Removing duplicates and rebalancing classes not only enhanced fairness but also improved the quality and focus of saliency visualizations.
3. **Attention does not guarantee interpretability.**
   Adding redundant attention layers to SE-based models introduces unnecessary complexity and may reduce transparency.
4. **Regularization can weaken human-understandable reasoning.**
   Label smoothing demonstrated that higher accuracy does not necessarily correspond to more meaningful explanations.
5. **Explainability must be evaluated alongside accuracy.**
   High test accuracy is only clinically valuable when supported by consistent, anatomically valid activation patterns.

Ultimately, the study shows that balanced optimization, careful preprocessing, and consistent interpretability validation are more effective for trustworthy medical AI than purely architectural innovations. The final optimized EfficientNetV2-S model achieving 0.989 accuracy with the clearest XAI maps embodies this principle by combining quantitative excellence with qualitative transparency.

6. Conclusion and Future Work

6.1 Conclusion

This thesis presented an optimized and interpretable deep learning framework for brain tumor classification using the EfficientNetV2-S architecture. The study systematically examined how optimization strategies, data preprocessing, and regularization affect both model performance and explainability in MRI-based medical diagnosis.

The proposed pipeline achieved a test accuracy of 0.989, combining high predictive power with clinically meaningful interpretability. Through a structured series of experiments, the research demonstrated that:

1. **Optimization and data integrity outweigh architectural complexity.**
   The AdamW optimizer and controlled augmentation achieved stable convergence and balanced recall across all tumor types, outperforming attention-based variants.
2. **Attention mechanisms are not universally beneficial.**
   Although theoretically advantageous, ECA, GAM, and CBAM introduced redundancy and slightly reduced performance, indicating that EfficientNetV2-S's internal SE blocks already provide sufficient feature recalibration.
3. **Improved preprocessing enhances fairness and interpretability.**
   The removal of 426 duplicate images, balanced augmentation, and moderate brightness normalization resulted in cleaner saliency maps and more consistent predictions, proving that data quality directly impacts model transparency.
4. **Label smoothing illustrates the accuracy–interpretability trade-off.**
   Although it slightly improved test accuracy to 0.990, it produced less focused activation regions, reducing the clarity of the model's visual reasoning.
5. **Explainability validation is essential for trustworthy AI.**
   Multi-method XAI analysis (Grad-CAM, Grad-CAM++, Score-CAM, SmoothGrad) confirmed that the optimized model's decisions aligned closely with true tumor regions, satisfying the core requirement of interpretability in clinical AI.

Together, these findings underscore a fundamental insight:
achieving state-of-the-art performance in medical imaging is not solely a matter of network design, but rather of responsible optimization, data hygiene, and interpretability validation. The proposed EfficientNetV2-S pipeline provides a strong foundation for developing transparent, reliable, and clinically deployable diagnostic models.

6.2 Future Work

Although the current framework achieved strong results, several directions remain open for future research and clinical translation:

1. **3D MRI Integration:**
   Extend the approach to volumetric 3D CNNs that can exploit inter-slice spatial correlations, improving lesion characterization and volumetric segmentation accuracy.
2. **Advanced Explainability (LIME, SHAP, and Concept Activation):**
   Incorporate model-agnostic explainers such as LIME and SHAP to complement CAM-based heatmaps with global feature attribution and human-interpretable concept quantification.
3. **Radiologist-in-the-Loop Validation:**
   Conduct user studies with clinical experts to evaluate whether XAI visualizations assist in real diagnostic workflows, measuring interpretability from a human-centered perspective.
4. **Adversarial Robustness and Trust Calibration:**
   Investigate how noise, artifacts, and adversarial perturbations affect both prediction and explanation stability, aiming to develop more resilient medical AI systems.
5. **Cross-Dataset Generalization:**
   Test the model on independent MRI datasets from different hospitals or imaging protocols to validate external robustness and reduce domain dependency.

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